



Sumário

Data Structures	1
Dsu	1
Mo	1
Fenwick Tree	2
Fenwick Tree 2D	2
Dynamic Fenwick Tree 2D	2
Sparse Table	2
Disjoint Sparse Table	3
Segment Tree	3
Lazy Segment Tree	3
Dual Segment Tree	4
Segment Tree Beats	4
Dynamic Segment Tree	5
Dynamic Persistent Segment Tree	5
Line Container (CHT)	5
Graphs	6
Dijkstra	6
MST (Kruskal)	6
Euler Path	6
Lca Sparse Table	6
Lca Binary Lifting	6
Tarjan	7
Articulations	7
Bridges	7
2-SAT	7
Hld Commutative	8
Hld Commutative Lazy	8
Hld Non Commutative	8
Hld Non Commutative Lazy	8
Dinic	9
Min Cost Max Flow	10
Bipartite Matching	10
Math	10
Mint	10
Fast Exponentiation	11
Sieve	11
Euler Phi	11
Number Theory	11
Miller Rabin	12
Matrix	12
Gaussian Elimination	12
Linear Basis	12
Unimodal Search	13
FFT	13
FFT Mod	13
NTT	13
FWHT	14
String	14
KMP	14
Z Function	14
String Hash	14
Trie	15
Manacher	15
Suffix Array (NLogN)	15
Suffix Array (Linear)	16
Geometry	16
Point	16
Line	17
Circle	17
Polygon	17
Convex Hull	18
Closest Pair	18
Others	18
Stress Test	18
Gen	18
Checker	18
Hash Function	18

Hash Verifier	18
Tabelas de Referência	20
Tipos, Constantes & Complexidade	20
Combinatória	20
Somatórios	20
Identidades Binomiais	20
Contagem	20
Sequências Especiais	20
Funções Geradoras	20
Potências Fatoriais & Bernoulli	20
Teoria dos Números	20
Euler Phi & Möbius	20
GCD, Diofantinas & TCR	20
Aritmetica Modular	20
Recorrências & Séries	20
Teorema Mestre	20
Séries de Taylor	20
Teoria dos Grafos	20
Teoremas	20
Tipos de Grafo	21
Matemática	21
Probabilidade	21
Trigonometria	21
Geometria	21
Teoria dos Jogos	21
Tabelas Numéricas	21
Números Primos	21
Fibonacci	21
Triângulo de Pascal $\binom{n}{k}$	21
Debugging	21

Data Structures

Dsu

```

/* DSU (Disjoint Set Union / Union-Find)
 * Mantem conjuntos disjuntos com uniao e busca eficiente.
 * Complexidade: O(a(N)); a e a inversa de Ackermann.
 * Memoria: O(N)
 * Metodos:
 * - find(i): Retorna o representante do conjunto de i.
 * - join(i, j): Une os conjuntos de i e j, retorna true se uniu.
 * - same(i, j): Verifica se i e j estao no mesmo conjunto.
 * - size(i): Retorna o tamanho do conjunto de i.
 */

```

```

D56 struct DSU {
1A8   int n;
923   vector<int> parent;
10F   vector<int> sz;
75E   int num_sets;
87D   DSU(int n) : n(n), parent(n), sz(n, 1), num_sets(n) {
4D6     iota(parent.begin(), parent.end(), 0);
3FC   }
C42   int find(int i) {
331     if (parent[i] == i) return i;
565     return parent[i] = find(parent[i]);
72C   }
D58   bool join(int i, int j) {
477     i = find(i), j = find(j);
41E     if (i != j) {
4C2       if (sz[i] < sz[j]) swap(i, j);
1F0       parent[j] = i;
529       sz[i] += sz[j];
CA8       num_sets--;
8A6       return true;
20F     }
D1F     return false;
30C   }
00C   int size(int i) { return sz[find(i)]; }
DA3   bool same(int i, int j) { return find(i) == find(j); }
EF5 };

```

Mo

```

/* Mo's Algorithm - O((N + Q) * sqrt(N))
 * queries: vetor de {l, r} 0-indexado, fechado.
 * add(i), rem(i): adicionam/removem o elemento i da janela.
 * ans(qi): registra resposta para a query de indice qi.
 *
 * Exemplo:
 *   vector<int> ans(q);
 *   Mo mo(n, queries);
 *   mo.run(
 *     [&](int i) { ... },
 *     [&](int i) { ... },
 *     [&](int qi) { ans[qi] = ...; }
 *   );
 */
D0F struct Mo {
7F4   int n, block;
670   struct Query {
738     int l, r, idx;
304   };
240   vector<Query> queries;
8CA   Mo(int n, vector<pair<int, int>> &qs)
10A     : n(n), block(max(1, (int)sqrt(n))) {
D26     for (int i = 0; i < (int)qs.size(); i++)
9AA       queries.push_back({qs[i].first, qs[i].second, i});
F94     sort(queries.begin(), queries.end(),
724       [&](const Query &a, const Query &b) {
ED6         int ba = a.l / block, bb = b.l / block;
A2D         if (ba != bb) return ba < bb;
FDf         return (ba & 1) ? a.r > b.r : a.r < b.r;
FC3       });
8F7   }
D5A   template <typename Add, typename Rem, typename Ans>
743   void run(Add add, Rem rem, Ans ans) {
73A     int cur_l = 0, cur_r = -1;
FFE     for (auto &[l, r, idx] : queries) {
90C       while (cur_r < r) add(++cur_r);
6A3       while (cur_l > l) add(--cur_l);
1B1       while (cur_r > r) rem(cur_r--);
E02       while (cur_l < l) rem(cur_l++);
414       ans(idx);
5B1     }
BB8   }
1FF };

/* HilbertMo - Mo com ordenacao pela curva de Hilbert
 * Mesma interface que Mo, mas com constante menor na pratica.
 */
CD0 struct HilbertMo {
670   struct Query {
738     int l, r, idx;
BFE     ll ord;
C8D   };
240   vector<Query> queries;
D16   static ll hilbert(int x, int y, int n) {
B72     ll d = 0;
F8A     for (int s = n >> 1; s; s >>= 1) {
7B9       int rx = (x & s) > 0, ry = (y & s) > 0;
71B       d += (ll)s * s * ((3 * rx) ^ ry);
069       if (!ry) {
1C6         if (rx) {
731           x = s - 1 - x;
D41           y = s - 1 - y;
1FD         }
9DD         swap(x, y);
FFF       }
0D1     }
BE2     return d;
416   }
E76   HilbertMo(int n, vector<pair<int, int>> &qs) {
ABE     int hn = 1;

```

```
FCC while (hn < n) hn <= 1;
D26 for (int i = 0; i < (int)qs.size(); i++)
082 queries.push_back(
A55 {qs[i].first, qs[i].second, i,
979 hilbert(qs[i].first, qs[i].second, hn)});
F94 sort(queries.begin(), queries.end(),
B18 [](const Query &a, const Query &b) {
73A return a.ord < b.ord;
A3A });
0D6 }

D5A template <typename Add, typename Rem, typename Ans>
743 void run(Add add, Rem rem, Ans ans) {
73A int cur_l = 0, cur_r = -1;
277 for (auto &[l, r, idx, ord] : queries) {
90C while (cur_r < r) add(++cur_r);
6A3 while (cur_l > l) add(--cur_l);
1B1 while (cur_r > r) rem(cur_r--);
E02 while (cur_l < l) rem(cur_l++);
414 ans(idx);
073 }
DC2 }
1AD };
```

Fenwick Tree

```
/* Fenwick Tree (Point Update, Prefix Query)
* Realiza atualizacoes pontuais e consultas de prefixo.
* Complexidade: O(log N) para update e query.
* Memoria: O(N)
* Requisitos:
* - NODE: operator+=, operator- e construtor default.
*/

/* --- Exemplo de NODE (Soma) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    void operator+=(const Node& other) { val += other.val; }
    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }
};
*/
```

```
8A0 template <typename NODE>
0F1 struct FenwickTree {
060 int N;
9B8 vector<NODE> bit;

5D2 FenwickTree(int n) : N(n), bit(n + 1) {}
D6B void update(int idx, NODE v) {
B2F for (idx++; idx <= N; idx += idx & -idx) {
046 bit[idx] += v;
23E }
9DA }

FCA NODE query(int idx) { // [0, idx]
ADA NODE res;
159 for (idx++; idx > 0; idx -= idx & -idx) {
8E6 res += bit[idx];
A38 }
B50 return res;
C00 }

762 NODE query(int l, int r) { // [l, r]
7C3 if (l > r) return NODE();
A4E return query(r) - query(l - 1);
821 }
134 };
```

Fenwick Tree 2D

```
/* Fenwick Tree 2D (Point Update, Rectangle Query)
* Atualizacoes pontuais e consultas de prefixo em matrizes.
* Complexidade: O(log N * log M) para update e query.
* Memoria: O(N * M)
* Requisitos:
```

```
* - NODE: operator+=, operator- e construtor default.
*/

/* --- Exemplo de NODE (Soma) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    void operator+=(const Node& other) { val += other.val; }
    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }
};
*/

8A0 template <typename NODE>
2E6 struct FenwickTree2D {
610 int N, M;
803 vector<vector<NODE>> bit;

265 FenwickTree2D(int n, int m)
B1E : N(n), M(m), bit(n + 1, vector<NODE>(m + 1)) {}

274 void update(int r, int c, NODE v) { // point update
3BA for (int i = r + 1; i <= N; i += i & -i) {
786 for (int j = c + 1; j <= M; j += j & -j) {
9ED bit[i][j] += v;
95F }
A1F }
BF9 }

82E NODE query(int r, int c) { // [(0,0), (r,c)]
ADA NODE res;
8D9 for (int i = r + 1; i > 0; i -= i & -i) {
83F for (int j = c + 1; j > 0; j -= j & -j) {
01D res += bit[i][j];
CB5 }
E20 }
B50 return res;
943 }

// Retangulo fechado [(r1,c1), (r2,c2)].
05F NODE query(int r1, int c1, int r2, int c2) {
5C4 if (r1 > r2 || c1 > c2) return NODE();
10A return query(r2, c2) - query(r1 - 1, c2) -
7F5 query(r2, c1 - 1) + query(r1 - 1, c1 - 1);
0E7 }
FF7 };
```

Dynamic Fenwick Tree 2D

```
/* Fenwick Tree 2D Sparse (Point Update, Rectangle Query)
* Suporta coordenadas ate 10^9 usando compressao interna.
* Complexidade: Build O(P log P), Update/Query O(log^2 P).
* Memoria: O(P log P).
* Requisitos:
* - NODE: operator+=, operator-, operator+ e construtor default.
* - Offline: passe todos os pontos de interesse no build.
*
* Creditos: Adaptado de TFG (Thiago Ferreira Goncalves)
* Fonte: github.com/tfg50/Competitive-Programming
*/
```

```
/* --- Exemplo de NODE (Soma com Inclusao-Exclusao) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    void operator+=(const Node& other) {
        val += other.val;
    }
    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }
    Node operator+(const Node& other) const {
        return Node(val + other.val);
    }
};
*/
```

```
8A0 template <typename NODE>
222 struct FenwickTree2DSparse {
6C1 vector<int> ord;
803 vector<vector<NODE>> bit;
E76 vector<vector<int>> coord;

BEF FenwickTree2DSparse(vector<pii> pts) {
CD7 sort(pts.begin(), pts.end());
5A8 for (auto [x, y] : pts) {
DC4 if (ord.empty() || x != ord.back()) ord.push_back(x);
ABA }

261 bit.resize(ord.size() + 1);
3EB coord.resize(ord.size() + 1);

A0A sort(pts.begin(), pts.end(), [](const pii &a, const pii &b) {
463 return a.second < b.second;
19C });

5A8 for (auto [x, y] : pts) {
9DE for (int i = get_x(x); i < bit.size(); i += i & -i) {
3E1 if (coord[i].empty() || coord[i].back() != y)
739 coord[i].push_back(y);
4F2 }
B39 }

8C2 for (int i = 0; i < bit.size(); i++)
F3E bit[i].assign(coord[i].size() + 1, NODE());
D9D }

7E8 int get_x(int x) {
2C8 return upper_bound(ord.begin(), ord.end(), x) - ord.begin();
95C }

D5E int get_y(int i, int y) {
E18 auto &c = coord[i];
BB0 return upper_bound(c.begin(), c.end(), y) - c.begin();
418 }

5CF void update(int x, int y, NODE v) {
9DE for (int i = get_x(x); i < bit.size(); i += i & -i) {
813 for (int j = get_y(i, y); j < bit[i].size(); j += j & -j) {
9ED bit[i][j] += v;
34E }
F86 }
390 }

84C NODE query(int x, int y) {
ADA NODE res;
FAD for (int i = get_x(x); i > 0; i -= i & -i) {
263 for (int j = get_y(i, y); j > 0; j -= j & -j) {
01D res += bit[i][j];
359 }
92F }
B50 return res;
7C4 }

FCA NODE query(int x1, int y1, int x2, int y2) {
276 return query(x2, y2) - query(x1 - 1, y2) -
12D query(x2, y1 - 1) + query(x1 - 1, y1 - 1);
944 }
B5F };
```

Sparse Table

```
/* Sparse Table (Static RMQ)
* Realiza consultas em range em tempo constante.
* Complexidade: O(N log N) no build, O(1) na query.
* Memoria: O(N log N)
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&) e
* construtor.
* - A operacao DEVE ser idempotente: f(a, a) = a.
*/

/* --- Exemplo de NODE (Indice do Minimo) ---
struct Node {
    int val, pos;
    Node(int v = 2e9, int p = -1) : val(v), pos(p) {}
    static Node merge(const Node& l, const Node& r) {
```

```

        return l.val <= r.val ? l : r;
    }
};

// Uso: SparseTable<Node> st(v) onde v e vector<Node>
// for (int i = 0; i < n; i++) {
//     cin >> x; v[i] = Node(x, i);
// }
// */

8A0 template <typename NODE>
7E9 struct SparseTable {
5F0     int N, K;
6F5     vector<vector<NODE>> st;
4A7     vector<int> lg;

67A     template <typename T>
E4A     SparseTable(const vector<T> &v) : N(v.size()) {
A77         K = (N > 0) ? __lg(N) : 0;
3A9         st.assign(K + 1, vector<NODE>(N));
641         lg.assign(N + 1, 0);
BB5         for (int i = 2; i <= N; i++) lg[i] = lg[i / 2] + 1;
DDD         for (int i = 0; i < N; i++) st[0][i] = NODE(v[i]);
2CE         for (int j = 1; j <= K; j++)
285             for (int i = 0; i + (1 << j) <= N; i++)
4E1                 st[j][i] = NODE::merge(st[j - 1][i],
A64                     st[j - 1][i + (1 << (j - 1))]);
CE2     }

762     NODE query(int l, int r) {
2C3         int j = lg[r - l + 1];
8F2         return NODE::merge(st[j][l], st[j][r - (1 << j) + 1]);
843     }
231 };

```

Disjoint Sparse Table

```

/* Disjoint Sparse Table (Static Range Query)
 * Query O(1) para qualquer operacao associativa.
 * Complexidade: O(N log N) no build, O(1) na query.
 * Memoria: O(N log N)
 * Requisitos:
 * - NODE deve ter: static merge(const NODE&, const NODE&) e
 * construtor.
 * - Operacao deve ser ASSOCIATIVA: f(f(a,b),c) =
 * f(a,f(b,c)).
 */

/* --- Exemplo de NODE (Soma em Range) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    static Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }
};
*/

8A0 template <typename NODE>
906 struct DisjointSparseTable {
5F0     int N, K;
6F5     vector<vector<NODE>> st;

67A     template <typename T>
68E     DisjointSparseTable(const vector<T> &v) : N(v.size()) {
3AA         if (N == 0) return;
475         K = __lg(2 * N - 1);
C78         st.assign(K, vector<NODE>(1 << K));
DDD         for (int i = 0; i < N; i++) st[0][i] = NODE(v[i]);
C54         for (int j = 1; j < K; j++) {
1C0             int len = 1 << j;
C2C             for (int m = len; m <= (1 << K); m += 2 * len) {
A34                 if (m < N) st[j][m] = st[0][m];
6D6                 for (int i = m + 1; i < min(N, m + len); i++)
0EA                     st[j][i] = NODE::merge(st[j][i - 1], st[0][i]);
C2E                 if (m - 1 < N) st[j][m - 1] = st[0][m - 1];
226                 for (int i = m - 2; i >= m - len; i--)

```

```

604         st[j][i] = NODE::merge(st[0][i], st[j][i + 1]);
643     }
3F0 }
4D7 }

762     NODE query(int l, int r) {
434         if (l == r) return st[0][l];
B89         int j = __lg(l ^ r);
894         return NODE::merge(st[j][l], st[j][r]);
33A     }
04C };

```

Segment Tree

```

/* Segment Tree (Point Update, Range Query)
 * Complexidade: O(log N) para update e query.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(V), e
 * construtor identidade.
 */

/* --- Exemplo de NODE (Soma com Update de Substituicao) ---
struct Node {
    ll val = 0;
    Node(ll v = 0) : val(v) {}

    static Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }

    // Para Point Set: val = v;
    // Para Point Add: val += v;
    void apply(ll v) {
        val = v;
    }
};
*/

8A0 template <typename NODE>
383 struct SegTree {
060     int N;
B1C     vector<NODE> seg;

FDE     explicit SegTree(int n) : N(n), seg(4 * n) {}

67A     template <typename T>
5DE     SegTree(const vector<T> &v) : SegTree((int)v.size()) {
231         build(1, 0, N - 1, v);
73D     }

67A     template <typename T>
5B9     void build(int no, int l, int r, const vector<T> &v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }

637     template <typename V>
AF7     void update(int no, int l, int r, int idx, V val) {
893         if (l == r) {
D88             seg[no].apply(val);
505             return;
EC9         }
27E         int m = (l + r) >> 1;
B73         if (idx <= m) update(no << 1, l, m, idx, val);
99C         else update((no << 1) | 1, m + 1, r, idx, val);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
7BC     }

38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << 1, l, m, a, b),

```

```

074         query((no << 1) | 1, m + 1, r, a, b));
365     }

637     template <typename V>
864     void update(int idx, V val) {
E1B         update(1, 0, N - 1, idx, val);
22F     }
36F     NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
0C7 };

```

Lazy Segment Tree

```

/* Lazy Segment Tree (Range Update, Range Query)
 * Realiza atualizacoes e consultas em range.
 * Complexidade: O(log N) para update e query.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(TAG, L, R) e
 * construtor identidade.
 * - TAG deve ter: compose(TAG) e construtor identidade.
 */

/* --- Exemplo de NODE e TAG (Soma e Multiplicacao com
 * Modulo) ---
struct Tag {
    ll mul = 1, add = 0;
    void compose(const Tag& t) {
        add = (add * t.mul + t.add) % MOD;
        mul = (mul * t.mul) % MOD;
    }
};

struct Node {
    ll val = 0;
    Node(ll v = 0) : val(v) {}
    static Node merge(const Node& l, const Node& r) {
        return Node((l.val + r.val) % MOD);
    }
    void apply(const Tag& t, int l, int r) {
        val = (val * t.mul + t.add * (r - l + 1)) % MOD;
    }
};
*/

708 template <typename NODE, typename TAG>
2BA struct LazySegmentTree {
060     int N;
B1C     vector<NODE> seg;
71A     vector<TAG> lazy;

BB8     explicit LazySegmentTree(int n)
63C         : N(n), seg(4 * n), lazy(4 * n) {}

67A     template <typename T>
18A     LazySegmentTree(const vector<T> &v)
002         : LazySegmentTree((int)v.size()) {
231         build(1, 0, N - 1, v);
9BB     }

67A     template <typename T>
5B9     void build(int no, int l, int r, const vector<T> &v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }

F1A     void push(int no, int lo, int hi) {
946         int m = (lo + hi) >> 1;
8F3         int l = no << 1, r = l | 1;
D09         seg[l].apply(lazy[no], lo, m);
9FE         lazy[l].compose(lazy[no]);
204         seg[r].apply(lazy[no], m + 1, hi);
3FC         lazy[r].compose(lazy[no]);

```

```

47F lazy[no] = TAG();
DD1 }

130 void update(int no, int l, int r, int a, int b, const TAG &v) {
2E6     if (b < l || r < a) return;
02B     if (a <= l && r <= b) {
8C7         seg[no].apply(v, l, r);
92C         lazy[no].compose(v);
505         return;
81F     }
FF9     push(no, l, r);
27E     int m = (l + r) >> 1;
9AB     update(no << l, l, m, a, b, v);
087     update((no << 1) | 1, m + 1, r, a, b, v);
DEB     seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
A1A }

38D NODE query(int no, int l, int r, int a, int b) {
2E6     if (b < l || r < a) return NODE();
83F     if (a <= l && r <= b) return seg[no];
FF9     push(no, l, r);
27E     int m = (l + r) >> 1;
AFE     return NODE::merge(query(no << l, l, m, a, b),
074                             query((no << 1) | 1, m + 1, r, a, b));
80A }

4E7 void update(int l, int r, const TAG &v) {
E27     update(1, 0, N - 1, l, r, v);
0D7 }

36F NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
3CE };

```

Dual Segment Tree

```

/* Dual Segment Tree (Range Update, Point Query)
 * Realiza atualizacoes em range e consultas pontuais.
 * Complexidade: O(log N) para update e get.
 * Memoria: O(4*N)
 * Requisitos:
 * - TAG deve ter: compose(TAG), apply(T&) e construtor
 * identidade.
 * - T e o tipo do dado nas folhas.
 * Exemplo de uso: DualSegTree<ll, MyTag> st(v);
 */

/* --- Exemplo de TAG (Soma e Multiplicacao com Modulo) ---
struct Tag {
    ll mul = 1, add = 0;
    void compose(const Tag& t) {
        mul = (mul * t.mul) % MOD;
        add = (add * t.mul + t.add) % MOD;
    }
    void apply(ll& val) { val = (val * mul + add) % MOD; }
};
*/

2E5 template <typename T, typename TAG>
9C1 struct DualSegTree {
060     int N;
71A     vector<TAG> lazy;
E4B     vector<T> leaves;

F06     explicit DualSegTree(int n) : N(n), lazy(4 * n), leaves(n) {}
5F6     DualSegTree(const vector<T> &v)
CE6         : N((int)v.size()), lazy(4 * v.size()), leaves(v) {}

78E     void push(int no) {
8A1         int l = no << 1;
C9C         int r = l | 1;
9FE         lazy[l].compose(lazy[no]);
3FC         lazy[r].compose(lazy[no]);
47F         lazy[no] = TAG();
283     }

130     void update(int no, int l, int r, int a, int b, const TAG &v) {
2E6         if (b < l || r < a) return;
02B         if (a <= l && r <= b) {
92C             lazy[no].compose(v);

```

```

505         return;
CA0     }
523     push(no);
27E     int m = (l + r) >> 1;
9AB     update(no << l, l, m, a, b, v);
087     update((no << 1) | 1, m + 1, r, a, b, v);
BEB }

629 T get(int no, int l, int r, int idx) {
893     if (l == r) {
4C5         T res = leaves[idx];
2E3         lazy[no].apply(res);
B50         return res;
BE7     }
523     push(no);
27E     int m = (l + r) >> 1;
52E     if (idx <= m) return get(no << l, l, m, idx);
483     else return get((no << 1) | 1, m + 1, r, idx);
C4A }

C8C void update(int l, int r, const TAG &t) {
56D     update(1, 0, N - 1, l, r, t);
A21 }
AE4 T get(int idx) { return get(1, 0, N - 1, idx); }
25C };

```

Segment Tree Beats

```

/* Segment Tree Beats (Range Update, Range Query)
 * Lazy Seg Tree com condicoes de parada e aplicacao de tag.
 * Complexidade amortizada: O(N log² N) por sequencia de updates.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(TAG, L, R),
 * break_condition(TAG) e tag_condition(TAG),
 * e construtor identidade.
 * - TAG deve ter: compose(TAG) e construtor identidade.
 *
 * break_condition(tag): ignora update ja satisfeito (mx <= v).
 * tag_condition(tag): aplica sem descer (mx2 < v <= mx).
 */

/* --- Exemplo de NODE e TAG (chmin em range + query de
 * soma/max) ---
struct Tag {
    ll val = LLONG_MAX; // identidade: nao faz nada
    void compose(const Tag& t) {
        val = min(val, t.val);
    }
};

struct Node {
    ll sum, mx, mx2;
    int cnt_mx, sz;

    Node()
        : sum(0), mx(LLONG_MIN), mx2(LLONG_MIN),
        cnt_mx(0), sz(0) {}

    Node(ll v)
        : sum(v), mx(v), mx2(LLONG_MIN), cnt_mx(1),
        sz(1) {}

    static Node merge(const Node& l, const Node& r) {
        Node res;
        res.sz = l.sz + r.sz;
        res.sum = l.sum + r.sum;
        if (l.mx == r.mx) {
            res.mx = l.mx; res.cnt_mx = l.cnt_mx + r.cnt_mx;
            res.mx2 = max(l.mx2, r.mx2);
        } else if (l.mx > r.mx) {
            res.mx = l.mx; res.cnt_mx = l.cnt_mx;
            res.mx2 = max(l.mx2, r.mx);
        } else {
            res.mx = r.mx; res.cnt_mx = r.cnt_mx;
            res.mx2 = max(l.mx, r.mx2);
        }
        return res;
    }
};

```

```

// aplica a tag quando tag_condition e verdadeiro
void apply(const Tag& t, int l, int r) {
    sum -= (ll)(mx - t.val) * cnt_mx;
    mx = t.val;
}

// ignora o update completamente (no ja satisfaz)
bool break_condition(const Tag& t) const {
    return mx <= t.val;
}

// pode aplicar a tag diretamente sem descer
bool tag_condition(const Tag& t) const {
    return mx2 < t.val;
}

};
*/

708 template <typename NODE, typename TAG>
BC8 struct SegTreeBeats {
060     int N;
B1C     vector<NODE> seg;
71A     vector<TAG> lazy;

249     explicit SegTreeBeats(int n) : N(n), seg(4 * n), lazy(4 * n) {}

67A     template <typename T>
47B     SegTreeBeats(const vector<T> &v)
001         : SegTreeBeats((int)v.size()) {
231         build(1, 0, N - 1, v);
1EF     }

67A     template <typename T>
5B9     void build(int no, int l, int r, const vector<T> &v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << l, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }

F1A     void push(int no, int lo, int hi) {
946         int m = (lo + hi) >> 1;
8F3         int l = no << 1, r = l | 1;
0D9         seg[l].apply(lazy[no], lo, m);
9FE         lazy[l].compose(lazy[no]);

204         seg[r].apply(lazy[no], m + 1, hi);
3FC         lazy[r].compose(lazy[no]);

47F         lazy[no] = TAG();
DD1     }

130     void update(int no, int l, int r, int a, int b, const TAG &v) {
FBB         if (b < l || r < a || seg[no].break_condition(v)) return;
40D         if (a <= l && r <= b && seg[no].tag_condition(v)) {
8C7             seg[no].apply(v, l, r);
92C             lazy[no].compose(v);
505             return;
2B2         }
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
9AB         update(no << l, l, m, a, b, v);
087         update((no << 1) | 1, m + 1, r, a, b, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
18B     }

38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << l, l, m, a, b),
074                                 query((no << 1) | 1, m + 1, r, a, b));
80A     }

```

```

4E7 void update(int l, int r, const TAG &v) {
E27     update(l, 0, N - 1, l, r, v);
0D7 }
36F NODE query(int l, int r) { return query(l, 0, N - 1, l, r); }
F92 };

```

Dynamic Segment Tree

```

/* Dynamic Segment Tree + Lazy (Range Update, Range Query)
 * Utilizada para intervalos gigantes (ex: 0 a 10^18).
 * Complexidade: O(log N) por operacao.
 * Memoria: O(Q log N) onde Q e o numero de operacoes.
 * Requisitos:
 * - NODE deve ter: static merge(const NODE&, const NODE&),
 *   apply(const TAG&, ll, ll) e construtor identidade.
 * - TAG deve ter: apply(const TAG&), campo 'has' (bool) e
 *   construtor identidade.
 */

/* --- Exemplo de NODE e TAG ---
struct Tag {
    ll add = 0;
    bool has = false;
    void apply(const Tag& t) {
        add += t.add;
        has = true;
    }
};

struct Node {
    ll val = 0;
    static Node merge(const Node& l, const Node& r) {
        return {l.val + r.val};
    }
    void apply(const Tag& t, ll l, ll r) {
        val += t.add * (r - l + 1);
    }
};
*/

708 template <typename NODE, typename TAG>
77D struct DynamicSegTree {
126     struct InternalNode {
4EB         NODE node;
657         TAG tag;
74F         int l, r;
FB6         InternalNode() : node(NODE()), tag(TAG()), l(-1), r(-1) {}

A61         void apply(const TAG &t, ll l, ll r) {
BE6             node.apply(t, l, r);
D21             tag.apply(t);
9C3         }
2FB     };

9B3     ll L, R;
4C1     vector<InternalNode> seg;

5C7     DynamicSegTree(ll l, ll r) : L(l), R(r) {
7A0         seg.reserve(4e6);
23E         seg.emplace_back();
BFD     }

4F9     void push(int v, ll l, ll r) {
A16         if (!seg[v].tag.has) return;
56B         if (l == r) return seg[v].tag = TAG();

4D5         if (seg[v].l == -1) {
9EF             seg[v].l = seg.size();
23E             seg.emplace_back();
579         }

CF3         if (seg[v].r == -1) {
90C             seg[v].r = seg.size();
23E             seg.emplace_back();
203         }

769         ll mid = l + (r - l) / 2;
F3E         seg[seg[v].l].apply(seg[v].tag, l, mid);
0B7         seg[seg[v].r].apply(seg[v].tag, mid + 1, r);
DD0         seg[v].tag = TAG();

```

```

AD7     }

352     void update(int v, ll l, ll r, ll ql, ll qr, const TAG &t) {
151         if (ql <= l && r <= qr) return seg[v].apply(t, l, r);

A48         push(v, l, r);
769         ll mid = l + (r - l) / 2;
441         if (ql <= mid) {
4D5             if (seg[v].l == -1) {
9EF                 seg[v].l = seg.size();
23E                 seg.emplace_back();
579             }
2DD             update(seg[v].l, l, mid, ql, qr, t);
0B1         }
64D         if (qr > mid) {
CF3             if (seg[v].r == -1) {
90C                 seg[v].r = seg.size();
23E                 seg.emplace_back();
203             }
76B             update(seg[v].r, mid + 1, r, ql, qr, t);
216         }

671         NODE lc = (seg[v].l != -1) ? seg[seg[v].l].node : NODE();
8A0         NODE rc = (seg[v].r != -1) ? seg[seg[v].r].node : NODE();
37A         seg[v].node = NODE::merge(lc, rc);
BF0     }

EE0     NODE query(int v, ll l, ll r, ll ql, ll qr) {
5CF         if (v == -1 || ql > r || qr < l) return NODE();
3C9         if (ql <= l && r <= qr) return seg[v].node;
A48         push(v, l, r);
769         ll mid = l + (r - l) / 2;
211         return NODE::merge(query(seg[v].l, l, mid, ql, qr),
0CC             query(seg[v].r, mid + 1, r, ql, qr));
933     }

FEE     void update(ll ql, ll qr, const TAG &t) {
B79         update(0, L, R, ql, qr, t);
4CA     }

323     NODE query(ll ql, ll qr) { return query(0, L, R, ql, qr); }
DBC };

```

Dynamic Persistent Segment Tree

```

/* Persistent Dynamic Segment Tree (Point Update, Range Query)
 * Consulta versoes anteriores e aloca nos sob demanda.
 * Ideal para intervalos grandes e multiplas versoes.
 * Complexidade: O(log N) por update e query.
 * Memoria: O(Q log N) onde Q e o numero de updates.
 * Requisitos:
 * - NODE deve ter: static merge(const NODE&, const NODE&) e
 *   construtor identidade.
 */

/* --- Exemplo de NODE (Soma) ---
struct Node {
    int val;
    Node(int v = 0) : val(v) {}
    static Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }
};
*/

8A0 template <typename NODE>
68A struct PersistentSegmentTree {
126     struct InternalNode {
1DF         NODE data;
74F         int l, r;
E61         InternalNode() : data(NODE()), l(0), r(0) {}
B12     };

060     int N;
788     vector<InternalNode> st;
34F     vector<int> roots;

BC9     PersistentSegmentTree(int n) : N(n) {
4E7         st.reserve(4e6);
F96         st.emplace_back();

```

```

405     roots.push_back(0);
71E     }

D69     int update(int root, int L, int R, int idx, NODE v) {
4C0         int node = st.size();
DF4         if (root == 0) st.emplace_back();
8FC         else st.push_back({st[root]});

651         if (L == R) {
DC5             st[node].data = NODE::merge(st[node].data, v);
192             return node;
BE8         }

08E         int mid = L + (R - L) / 2;

// A ordem do LHS importa antes do C++17.
D33         if (idx <= mid) {
561             int prev = (root == 0) ? 0 : st[root].l;
339             st[node].l = update(prev, L, mid, idx, v);
413         } else {
AFD             int prev = (root == 0) ? 0 : st[root].r;
674             st[node].r = update(prev, mid + 1, R, idx, v);
066         }

98D         auto& nd = st[node];
0EB         nd.data = NODE::merge(st[nd.l].data, st[nd.r].data);
192         return node;
60F     }

082     NODE query(int node, int L, int R, int i, int j) {
0B8         if (node == 0 || i > R || j < L) return NODE();
692         if (i <= L && R <= j) return st[node].data;
08E         int mid = L + (R - L) / 2;
B1E         return NODE::merge(query(st[node].l, L, mid, i, j),
521             query(st[node].r, mid + 1, R, i, j));
26E     }

D6B     void update(int idx, NODE v) {
69E         roots.push_back(update(roots.back(), 0, N - 1, idx, v));
1BF     }

3A5     NODE query(int version, int l, int r) {
191         return query(roots[version], 0, N - 1, l, r);
247     }
B56 };

```

Line Container (CHT)

```

/* Convex Hull Trick dinamico - O(log N) por operacao
 *
 * Mantem retas y = k*x + m e responde o MAXIMO em x.
 * Aceita k e x arbitrarios; nao exige monotonicidade.
 *
 * Uso: DP dp[i] = max_j(k[j].x[i] + m[j]). Cada estado vira
 * add(k[j], m[j]); a transicao vira query(x[i]).
 *
 * add(k, m) → insere a reta y = k*x + m
 * query(x) → maior valor entre todas as retas em x
 *
 * Para MÍNIMO: insira add(-k, -m) e use -query(x).
 */

72C struct Line {
F3E     ll k, m;
6F3     mutable ll p;
CA5     bool operator<(const Line &o) const { return k < o.k; }
ABF     bool operator<(ll x) const { return p < x; }
868 };

781 struct LineContainer : multiset<Line, less<>> {
FD2     static const ll inf = LLONG_MAX;
10F     ll div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a % b); }
A1C     bool isect(iterator x, iterator y) {
A95         if (y == end()) return x->p = inf, 0;
9CB         if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
591         else x->p = div(y->m - x->m, x->k - y->k);
870         return x->p >= y->p;
2FA     }

A0C     void add(ll k, ll m) {
116         auto z = insert({k, m, 0}), y = z++, x = y;

```

```

7B1  while (isect(y, z)) z = erase(z);
141  if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
57D  while ((y = x) != begin() && (--x)->p >= y->p)
774    isect(x, erase(y));
086  }
4AD  ll query(ll x) {
229    assert(!empty());
7D1    auto l = *lower_bound(x);
96A    return l.k * x + l.m;
D21  }
577  };

```

Graphs

Dijkstra

```

/* Dijkstra – Caminho minimo de fonte unica
* Complexidade:  $O((V + E) \log V)$ 
* Requer pesos nao-negativos.
*
* adj[u] = lista de {v, w}: aresta u-v com peso w
*
* dist[v] = distancia minima de s ate v.
* dist[v] = INF se v e inalcançavel.
*/

67A template <typename T>
5D5 vector<T> dijkstra(const vector<vector<pair<int, T>>> &adj,
E60 int s) {
B4C  int n = adj.size();
305  const T INF = numeric_limits<T>::max();
835  vector<T> dist(n, INF);
69C  priority_queue<pair<T, int>, vector<pair<T, int>>, greater<>>
018  pq;
A93  dist[s] = 0;
7BA  pq.push({0, s});
502  while (!pq.empty()) {
2F9    auto [d, u] = pq.top();
716    pq.pop();
3E1    if (d > dist[u]) continue;
610    for (auto [v, w] : adj[u])
DDE      if (dist[u] + w < dist[v]) {
491        dist[v] = dist[u] + w;
BF3        pq.push({dist[v], v});
D74      }
9AE    }
8D7  return dist;
6D3 }

```

MST (Kruskal)

```

32D #include "dsu.hpp"

/* MST – Arvore Geradora Minima (Kruskal + DSU)
* Complexidade:  $O(E \log E)$ . Grafo nao-direcionado ponderado.
*
* mst(n, edges) → {custo total, arestas usadas}
* edges: vetor de {peso, u, v}.
* Se desconexo, devolve a floresta minima; used tem < n-1.
*
* Para arvore MAXIMA, negue os pesos ou ordene decrescente.
*/

67A template <typename T>
FD5 pair<T, vector<pair<int, int>>>
652 mst(int n, vector<tuple<T, int, int>> edges) {
FF1  sort(edges.begin(), edges.end());
370  DSU dsu(n);
C80  T total = 0;
24A  vector<pair<int, int>> used;
DC1  for (auto &[w, u, v] : edges)
5DF    if (dsu.join(u, v)) {
532      total += w;
12C      used.push_back({u, v});
88B    }
0CF  return {total, used};

```

```

290 }

```

Euler Path

```

/* Caminho/Circuito Euleriano – Hierholzer –  $O(V + E)$ 
*
* Condiçoes de existencia:
* Grafo nao-direcionado:
* - Circuito: todos os vertices tem grau par.
* - Caminho: exatamente dois tem grau impar.
* Grafo direcionado:
* - Circuito: in_degree[v] == out_degree[v] para todo v.
* - Caminho: um vertice com out-in=1, um com in-out=1.
*
* euler(adj, s) → caminho com E+1 vertices; vazio se invalido.
* adj[u] = lista de {v, edge_id} para arestas u-v.
* No nao-direcionado, insira ambas as direcoes com o mesmo id.
*/

E48 vector<int> euler(vector<vector<pair<int, int>>> &adj, int s) {
CE9  int edges = 0;
14F  for (auto &v : adj) edges += v.size();
// Para direcionado, remova a divisao por 2.
2D5  edges /= 2;

B27  vector<bool> used(edges, false);
8E1  vector<int> ptr(adj.size(), 0);
DAB  vector<int> path, stk = {s};

07D  while (!stk.empty()) {
9E9    int v = stk.back();
2C4    bool found = false;
EBE    while (ptr[v] < (int)adj[v].size()) {
FFC      auto [u, eid] = adj[v][ptr[v]++];
AC3      if (!used[eid]) {
C3D        used[eid] = true;
967        stk.push_back(u);
E96        found = true;
C2B        break;
BE7      }
7CB    }
8C1    if (!found) {
80A      path.push_back(v);
518      stk.pop_back();
9D9    }
8D6  }

66D  if ((int)path.size() != edges + 1) return {};
3A9  reverse(path.begin(), path.end());
535  return path;
924 }

```

Lca Sparse Table

```

D7B #include "sparse-table.hpp"

/* LCA (Lowest Common Ancestor) via RMQ
* Ancestral comum mais proximo de dois vertices da arvore.
* Complexidade:  $O(N \log N)$  no build,  $O(1)$  por query.
* Memoria:  $O(N \log N)$ 
* Requisitos:
* - Grafo deve ser uma arvore (conexo e aciclico).
* Metodos:
* - lca(u, v): Retorna o LCA de u e v.
* - dist(u, v): Retorna a distancia entre u e v.
* - is_ancestor(u, v): Verifica se u e ancestral de v.
* - parent(u): Retorna o pai de u.
*/

542 struct LCANode {
43D  int dep, id;
89D  LCANode(int d = 1e9, int pos = -1) : dep(d), id(pos) {}
3E4  static LCANode merge(const LCANode &l, const LCANode &r) {
DC4    return l.dep < r.dep ? l : r;
0A6  }
008 };

33E struct LCA {

```

```

060 int N;
41B vector<int> first, tour, d, p;
73B SparseTable<LCANode> st;

DFC  LCA(const vector<vector<int>> &adj, int root = 0)
B9F    : N(adj.size()), first(N), d(N), p(N, -1),
CDD    st(vector<LCANode>()) {

AA1    tour.reserve(2 * N);
52E    vector<LCANode> nodes;
A7F    nodes.reserve(2 * N);

70E    auto dfs = [&](auto self, int u, int par, int dep) -> void {
9A7      p[u] = par;
701      d[u] = dep;
D86      first[u] = tour.size();
FD8      tour.push_back(u);
612      nodes.push_back({dep, (int)tour.size() - 1});
372      for (int v : adj[u]) {
D56        if (v == par) continue;
207        self(self, v, u, dep + 1);
FD8        tour.push_back(u);
612        nodes.push_back({dep, (int)tour.size() - 1});
B41      }
958    };

7D9    dfs(dfs, root, -1, 0);
F2E    st = SparseTable<LCANode>(nodes);
490  }

47F  int parent(int u) { return p[u]; }
310  int lca(int u, int v) {
B63    int l = first[u], r = first[v];
A13    if (l > r) swap(l, r);
369    return tour[st.query(l, r).id];
1E6  }

C11  int dist(int u, int v) {
05C    return d[u] + d[v] - 2 * d[lca(u, v)];
465  }

C22  bool is_ancestor(int u, int v) { return lca(u, v) == u; }
19C };

```

Lca Binary Lifting

```

/* LCA (Lowest Common Ancestor) via Binary Lifting
* Ancestral comum mais proximo de dois vertices da arvore.
* Complexidade:  $O(N \log N)$  no build,  $O(\log N)$  por query.
* Memoria:  $O(N \log N)$ 
* Requisitos:
* - Grafo deve ser uma arvore (conexo e aciclico).
* Metodos:
* - lca(u, v): Retorna o LCA de u e v.
* - dist(u, v): Retorna a distancia entre u e v.
* - is_ancestor(u, v): Verifica se u e ancestral de v.
* - parent(u): Retorna o pai de u (up[u][0]).
* - kth_ancestor(u, k): k-esimo ancestral; -1 se nao existe.
*/

33E struct LCA {
8B7  int N, LOG;
5CE  vector<int> d;
5FE  vector<vector<int>> up;

DFC  LCA(const vector<vector<int>> &adj, int root = 0)
839    : N(adj.size()), LOG(__lg(N) + 1), d(N),
F1E    up(N, vector<int>(LOG + 1, -1)) {

91B    auto dfs = [&](auto self, int u, int p, int dep) -> void {
701      d[u] = dep;
F1D      up[u][0] = (p == -1) ? u : p;
B16      for (int j = 1; j <= LOG; j++)
4D5        up[u][j] = up[up[u][j - 1]][j - 1];
372      for (int v : adj[u]) {
730        if (v == p) continue;
207        self(self, v, u, dep + 1);
320      }
3B1    };

```

```

7D9     dfs(dfs, root, -1, 0);
5A6   }
579   int parent(int u) { return up[u][0] == u ? -1 : up[u][0]; }
6AE   int kth_ancestor(int u, int k) {
472     if (k > d[u]) return -1;
4C8     for (int j = 0; j <= LOG; j++)
518       if ((k >> j) & 1) u = up[u][j];
03F     return u;
51C   }
310   int lca(int u, int v) {
4F1     if (d[u] < d[v]) swap(u, v);
3B4     int diff = d[u] - d[v];
4C8     for (int j = 0; j <= LOG; j++)
122       if ((diff >> j) & 1) u = up[u][j];
60E     if (u == v) return u;
CAE     for (int j = LOG; j >= 0; j--)
76F       if (up[u][j] != up[v][j]) {
0EF         u = up[u][j];
E4F         v = up[v][j];
D97       }
C15     return up[u][0];
B04   }
C11   int dist(int u, int v) {
05C     return d[u] + d[v] - 2 * d[lca(u, v)];
465   }
C22   bool is_ancestor(int u, int v) { return lca(u, v) == u; }
E8E };

```

Tarjan

```

/* Tarjan SCC (Strongly Connected Components)
 * Complexidade: O(V + E)
 *
 * Campos publicos apos build():
 * comp[v] = SCC de v, em ordem topologica reversa
 * scc[i] = lista de vertices do i-esimo SCC
 * dag = grafo condensado sem arestas duplicadas
 * n_scc = numero de SCCs
 *
 * Se ha aresta SCC a → SCC b, entao comp[a] > comp[b].
 * Para ordem topologica direta, percorra n_scc-1 ... 0.
 */
FA3 struct Tarjan {
058   int n, n_scc = 0;
A41   vector<vector<int>> g, scc, dag;
F7A   vector<int> comp;
770   Tarjan(int n) : n(n), g(n), comp(n, -1) {}
224   Tarjan(const vector<vector<int>> &adj)
422     : n(adj.size()), g(adj), comp(adj.size(), -1) {
6F2     build();
35F   }
C2B   void add_edge(int u, int v) { g[u].push_back(v); }
0A8   void build() {
E7C     vector<int> low(n), num(n, -1), stk;
C05     vector<bool> on_stk(n, false);
2DC     int timer = 0;
36D     function<void(int)> dfs = [&](int v) {
006       low[v] = num[v] = timer++;
B00       stk.push_back(v);
48C       on_stk[v] = true;
E32       for (int u : g[v]) {
403         if (num[u] == -1) {
512           dfs(u);
C80           low[v] = min(low[v], low[u]);
F19           } else if (on_stk[u]) low[v] = min(low[v], num[u]);
38B         }
46E         if (low[v] == num[v]) {
861           scc.push_back({});
667           while (true) {
B01             int u = stk.back();

```

```

518     stk.pop_back();
635     on_stk[u] = false;
65B     comp[u] = n_scc;
588     scc.back().push_back(u);
163     if (u == v) break;
87F   }
385   n_scc++;
6CD   }
5A5   };
830   for (int i = 0; i < n; i++)
917     if (num[i] == -1) dfs(i);
9E4   dag.assign(n_scc, {});
19E   for (int u = 0; u < n; u++)
4FB     for (int v : g[u])
D67       if (comp[u] != comp[v]) dag[comp[u]].push_back(comp[v]);
404   for (auto &adj : dag) {
324     sort(adj.begin(), adj.end());
942     adj.erase(unique(adj.begin(), adj.end()), adj.end());
80E   }
BA2   }
6D7 };

```

Articulations

```

/* Pontos de Articulaçao – Tarjan
 * Complexidade: O(V + E)
 * Grafo nao-direcionado.
 *
 * Sua remocao aumenta o numero de componentes conexas.
 *
 * Campos publicos apos build():
 * is_ap[v] = true se v e ponto de articulacao
 * get() = lista dos pontos de articulacao
 */
E5A struct Articulations {
1A8   int n;
789   vector<vector<int>> g;
BC9   vector<bool> is_ap;
620   Articulations(int n) : n(n), g(n), is_ap(n, false) {}
6F5   Articulations(const vector<vector<int>> &adj)
FDA     : n(adj.size()), g(adj), is_ap(adj.size(), false) {
6F2     build();
B9D   }
58B   void add_edge(int u, int v) {
F2B     g[u].push_back(v);
4D6     g[v].push_back(u);
FFF   }
0A8   void build() {
AC3     vector<int> num(n, -1), low(n);
2DC     int timer = 0;
CF3     function<void(int, int)> dfs = [&](int v, int par) {
006       low[v] = num[v] = timer++;
E07       int children = 0;
E32       for (int u : g[v]) {
403         if (num[u] == -1) {
C5F           children++;
412           dfs(u, v);
C80           low[v] = min(low[v], low[u]);
369           if (par == -1 && children > 1) is_ap[v] = true;
1B3           if (par != -1 && low[u] >= num[v]) is_ap[v] = true;
709         } else if (u != par) {
2D3           low[v] = min(low[v], num[u]);
933         }
9F1       }
8CA     };
830     for (int i = 0; i < n; i++)
10B       if (num[i] == -1) dfs(i, -1);
DEE   }
// Lista dos pontos de articulacao.

```

```

D3D   vector<int> get() const {
02F     vector<int> res;
830     for (int i = 0; i < n; i++)
BA3       if (is_ap[i]) res.push_back(i);
B50     return res;
60A   }
A5D };

```

Bridges

```

/* Pontes (Bridges) – Tarjan
 * Complexidade: O(V + E)
 * Grafo nao-direcionado.
 *
 * Uma aresta e ponte se sua remocao desconecta o grafo.
 *
 * Campos publicos apos build():
 * bridges = lista de arestas {u, v} que sao pontes
 */
D44 struct Bridges {
1A8   int n;
789   vector<vector<int>> g;
0CC   vector<pair<int, int>> bridges;
B0A   Bridges(int n) : n(n), g(n) {}
C65   Bridges(const vector<vector<int>> &adj)
30E     : n(adj.size()), g(adj) {
6F2     build();
5F7   }
58B   void add_edge(int u, int v) {
F2B     g[u].push_back(v);
4D6     g[v].push_back(u);
FFF   }
0A8   void build() {
AC3     vector<int> num(n, -1), low(n);
2DC     int timer = 0;
CF3     function<void(int, int)> dfs = [&](int v, int par) {
006       low[v] = num[v] = timer++;
E32       for (int u : g[v]) {
403         if (num[u] == -1) {
412           dfs(u, v);
C80           low[v] = min(low[v], low[u]);
2FE           if (low[u] > num[v]) bridges.push_back({v, u});
FB4         } else if (u != par) {
2D3           low[v] = min(low[v], num[u]);
E66         }
7A3       }
938     };
830     for (int i = 0; i < n; i++)
10B       if (num[i] == -1) dfs(i, -1);
A6F   }
000 };

```

2-SAT

```

C31 #include "tarjan.hpp"
/* 2-SAT – Complexidade: O(V + E)
 *
 * Literal x: use i para x_i=true, ~i para x_i=false.
 *
 * add_impl(x, y) → x → y e contrapositiva ~y → ~x
 * add_or(x, y) → x ∨ y
 * add_xor(x, y) → exatamente um de x, y e true
 * add_iff(x, y) → x ↔ y
 * force(x) → x = true
 * solve() → {satisfativo, ans[]}; ans[i] = 0 ou 1
 */
D9D struct TwoSat {
1A8   int n;
0D7   Tarjan t;
DAB   vector<int> ans;
46B   TwoSat(int n) : n(n), t(2 * n), ans(n, -1) {}

```

```

// Converte i em 2*i e ~i em 2*i+1.
507 static int node(int x) { return x >= 0 ? 2 * x : -2 * x - 1; }

// implicacao x -> y (e contrapositiva ~y -> ~x)
74A void add_impl(int x, int y) {
D64     int nx = node(x), ny = node(y);
2B2     t.add_edge(nx, ny);
378     t.add_edge(ny ^ 1, nx ^ 1);
255 }

// x v y
F10 void add_or(int x, int y) { add_impl(~x, y); }

// x XOR y (exatamente um verdadeiro)
487 void add_xor(int x, int y) {
27A     add_or(x, y);
0E8     add_or(~x, ~y);
C02 }

// x ↔ y
1FF void add_iff(int x, int y) {
2D0     add_impl(x, y);
EA9     add_impl(~x, ~y);
721 }

// forca x = true
5F8 void force(int x) { add_impl(~x, x); }

A8E pair<bool, vector<int>> solve() {
2B0     t.build();
E1C     ans.assign(n, -1);
603     for (int i = 0; i < n; i++) {
EBA         if (t.comp[2 * i] == t.comp[2 * i + 1]) return {false, {}};
049         ans[i] = t.comp[2 * i] < t.comp[2 * i + 1] ? 1 : 0;
EF0     }
997     return {true, ans};
194 }
AC6 };

```

Hld Commutative

```
0FA #include "segment-tree.hpp"
```

```

/* HLD Comutativo (Update Pontual)
 * Para operacoes onde merge(A, B) = merge(B, A).
 */

```

```

8A0 template <typename NODE>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
115     SegTree<NODE> st;

AA9     HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C         int root = 0)
F92     : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
AC1         st(n) {

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
C2C             d[v] = d[u] + 1;
E42             p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;

```

```

27A         }
CAE     }
07D     }
A11 };

B9B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
113     };

A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);

759     vector<NODE> base(n);
852     for (int i = 0; i < n; i++) base[pos[i]] = vals[i];
E4E     st = SegTree<NODE>(base);
1E0 }

080 void update(int u, const NODE &val) { st.update(pos[u], val); }

0F8 NODE query_path(int u, int v) {
ADA     NODE res;
0E0     while (head[u] != head[v]) {
44B         if (d[head[u]] > d[head[v]]) swap(u, v);
A0D         res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025         v = p[head[v]];
41F     }
0E0     if (d[u] > d[v]) swap(u, v);
DC7     return NODE::merge(res, st.query(pos[u], pos[v]));
81B }

51A NODE query_subtree(int u) {
0CF     return st.query(pos[u], pos[u] + sz[u] - 1);
747 }
058 };

```

Hld Commutative Lazy

```
064 #include "lazy-segment-tree.hpp"
```

```

/* HLD Comutativo com Lazy Propagation
 * Para operacoes onde merge(A, B) = merge(B, A).
 * Updates e queries em caminhos e subarvores.
 */

```

```

708 template <typename NODE, typename TAG>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
FFF     LazySegmentTree<NODE, TAG> st;

AA9     HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C         int root = 0)
F92     : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
AC1         st(n) {

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
C2C             d[v] = d[u] + 1;
E42             p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {

```

```

F10         if (g[u][i] == best_v && i != 0) {
D76             swap(g[u][0], g[u][i]);
C2B             break;
27A         }
CAE     }
07D     }
A11 };

B9B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
113     };

A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);

759     vector<NODE> base(n);
852     for (int i = 0; i < n; i++) base[pos[i]] = vals[i];
9F8     st = LazySegmentTree<NODE, TAG>(base);
631 }

D7C void update_path(int u, int v, const TAG &tag) {
0E0     while (head[u] != head[v]) {
44B         if (d[head[u]] > d[head[v]]) swap(u, v);
EC8         st.update(pos[head[v]], pos[v], tag);
025         v = p[head[v]];
DD0     }
0E0     if (d[u] > d[v]) swap(u, v);
07A     st.update(pos[u], pos[v], tag);
656 }

0F8 NODE query_path(int u, int v) {
ADA     NODE res;
0E0     while (head[u] != head[v]) {
44B         if (d[head[u]] > d[head[v]]) swap(u, v);
A0D         res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025         v = p[head[v]];
41F     }
0E0     if (d[u] > d[v]) swap(u, v);
DC7     return NODE::merge(res, st.query(pos[u], pos[v]));
81B }

F81 void update_subtree(int u, const TAG &tag) {
675     st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D }

51A NODE query_subtree(int u) {
0CF     return st.query(pos[u], pos[u] + sz[u] - 1);
747 }
AA5 };

```

Hld Non Commutative

```
0FA #include "segment-tree.hpp"
```

```

/* HLD Nao-Comutativo (Update Pontual)
 * Para merge(A, B) != merge(B, A).
 * DoubleNode guarda as duas direcoes do merge.
 */

```

```

8A0 template <typename NODE>
0FF struct DoubleNode {
A30     NODE down, up;
F17     DoubleNode() : down(NODE()), up(NODE()) {}
77F     DoubleNode(const NODE &n) : down(n), up(n) {}
31A     DoubleNode(const NODE &d, const NODE &u) : down(d), up(u) {}

25A     static DoubleNode merge(const DoubleNode &l,
553                             const DoubleNode &r) {
7E0         return {NODE::merge(l.down, r.down),
5D3                     NODE::merge(r.up, l.up)};
063     }
FB7 };

```

```

8A0 template <typename NODE>
403 struct HLD {
577     int n, t;
256     vector<int> p, sz, d, head, pos, tour;
58B     SegTree<DoubleNode<NODE>> st;
AA9     HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C         int root = 0)
F92         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
007         tour(n), st(n) {
898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;
94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
C2C             d[v] = d[u] + 1;
E42             p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
07D         }
A11     };
89B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
6EA         tour[pos[u]] = u;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
54A     };
A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);
9FC     vector<DoubleNode<NODE>> base(n);
830     for (int i = 0; i < n; i++)
9A8         base[pos[i]] = DoubleNode<NODE>(vals[i]);
19C     st = SegTree<DoubleNode<NODE>>(base);
05A }
E1B void update(int u, const NODE &val) {
353     st.update(pos[u], DoubleNode<NODE>(val));
443 }
0F8 NODE query_path(int u, int v) {
716     NODE l, r;
0E0     while (head[u] != head[v]) {
BF1         if (d[head[u]] > d[head[v]]) {
0D0             l = NODE::merge(l, st.query(pos[head[u]], pos[u]).up);
139             u = p[head[u]];
A08         } else {
455             r = NODE::merge(st.query(pos[head[v]], pos[v]).down, r);
025             v = p[head[v]];
A09         }
02B     }
152     if (d[u] > d[v]) {
AF0         l = NODE::merge(l, st.query(pos[v], pos[u]).up);
EF6     } else {
5AF         r = NODE::merge(st.query(pos[u], pos[v]).down, r);
F3A     }
79F     return NODE::merge(l, r);
22A }
F81 void update_subtree(int u, const TAG &tag) {
675     st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D }
51A NODE query_subtree(int u) {
DE6     return st.query(pos[u], pos[u] + sz[u] - 1).down;
588 }
3DA };

```

```

F3A     }
79F     return NODE::merge(l, r);
22A }
51A NODE query_subtree(int u) {
DE6     return st.query(pos[u], pos[u] + sz[u] - 1).down;
588 }
45D };

```

Hld Non Commutative Lazy

```

064 #include "lazy-segment-tree.hpp"
/* HLD Nao-Comutativo com Lazy Propagation
 * Para merge(A, B) != merge(B, A), como matrizes.
 * DoubleNode guarda as ordens normal e invertida do merge.
 */
8A0 template <typename NODE>
0FF struct DoubleNode {
A30     NODE down, up;
F17     DoubleNode() : down(NODE()), up(NODE()) {}
77F     DoubleNode(const NODE &n) : down(n), up(n) {}
31A     DoubleNode(const NODE &d, const NODE &u) : down(d), up(u) {}
25A     static DoubleNode merge(const DoubleNode &l,
553                             const DoubleNode &r) {
7E0         return {NODE::merge(l.down, r.down),
5D3             NODE::merge(r.up, l.up)};
063     }
BBF     template <typename TAG>
3CC     void apply(const TAG &t, int l, int r) {
B08         down.apply(t, l, r);
2BF         up.apply(t, l, r);
8FB     }
204 };
708 template <typename NODE, typename TAG>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
C94     LazySegmentTree<DoubleNode<NODE>, TAG> st;
AA9     HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C         int root = 0)
F92         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
AC1         st(n) {
898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;
94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
C2C             d[v] = d[u] + 1;
E42             p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
07D         }
A11     };
89B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
408         for (int i = 0; i < (int)g[u].size(); i++) {
BB6             if (g[u][i] == best_v && i != 0) {
F10                 swap(g[u][0], g[u][i]);
D76                 break;
C2B             }
27A         }
CAE     }
07D     }
A11 };
89B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {

```

```

06E         int v = g[u][i];
567         if (v == p[u]) continue;
BD7         head[v] = (i == 0 ? head[u] : v);
033         self(self, v);
D08     }
113 };
A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);
9FC     vector<DoubleNode<NODE>> base(n);
830     for (int i = 0; i < n; i++)
9A8         base[pos[i]] = DoubleNode<NODE>(vals[i]);
54E     st = LazySegmentTree<DoubleNode<NODE>, TAG>(base);
716 }
D7C void update_path(int u, int v, const TAG &tag) {
0E0     while (head[u] != head[v]) {
BF1         if (d[head[u]] > d[head[v]]) {
D76             st.update(pos[head[u]], pos[u], tag);
139             u = p[head[u]];
D79         } else {
EC8             st.update(pos[head[v]], pos[v], tag);
025             v = p[head[v]];
E8B         }
15B     }
B83     if (d[u] > d[v]) st.update(pos[v], pos[u], tag);
052     else st.update(pos[u], pos[v], tag);
855 }
0F8 NODE query_path(int u, int v) {
716     NODE l, r;
0E0     while (head[u] != head[v]) {
BF1         if (d[head[u]] > d[head[v]]) {
0D0             l = NODE::merge(l, st.query(pos[head[u]], pos[u]).up);
139             u = p[head[u]];
A08         } else {
455             r = NODE::merge(st.query(pos[head[v]], pos[v]).down, r);
025             v = p[head[v]];
A09         }
02B     }
152     if (d[u] > d[v]) {
AF0         l = NODE::merge(l, st.query(pos[v], pos[u]).up);
EF6     } else {
5AF         r = NODE::merge(st.query(pos[u], pos[v]).down, r);
F3A     }
79F     return NODE::merge(l, r);
22A }
F81 void update_subtree(int u, const TAG &tag) {
675     st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D }
51A NODE query_subtree(int u) {
DE6     return st.query(pos[u], pos[u] + sz[u] - 1).down;
588 }
3DA };

```

Dinic

```

/* Dinic (Max Flow)
 * Complexidade: O(V²E) geral; O(EV) em grafos unitarios.
 * Memoria: O(V + E)
 *
 * Metodos:
 * add_edge(u, v, cap) → aresta u→v com capacidade cap
 * add_edge(u, v, cap, rcap) → define capacidade reversa
 * f_low(s, t) → fluxo maximo de s a t
 * min_cut(s) → lado alcancavel da fonte apos flow()
 */
67A template <typename T>
14D struct Dinic {
E9B     struct Edge {
C63         int to, rev;
4AE         T cap;
001     };

```

```

060 int N;
ED3 vector<vector<Edge>> g;
538 vector<int> level, iter;

241 Dinic(int n) : N(n), g(n), level(n), iter(n) {}

D8F void add_edge(int u, int v, T cap, T rcap = 0) {
25C     g[u].push_back({v, (int)g[v].size(), cap});
ACA     g[v].push_back({u, (int)g[u].size() - 1, rcap});
05B }

123 bool bfs(int s, int t) {
224     fill(level.begin(), level.end(), -1);
26A     queue<int> q;
EC8     level[s] = 0;
08B     q.push(s);
14D     while (!q.empty()) {
B1E         int v = q.front();
833         q.pop();
24E         for (auto &e : g[v])
466             if (e.cap > 0 && level[e.to] < 0) {
BD7                 level[e.to] = level[v] + 1;
6F4                 q.push(e.to);
A33             }
FF9     }
FCB     return level[t] >= 0;
56A }

D20 T dfs(int v, int t, T f) {
704     if (v == t) return f;
6E7     for (int &i = iter[v]; i < (int)g[v].size(); i++) {
63C         Edge &e = g[v][i];
2C2         if (e.cap > 0 && level[v] < level[e.to]) {
054             T d = dfs(e.to, t, min(f, e.cap));
1A3             if (d > 0) {
8FC                 e.cap -= d;
772                 g[e.to][e.rev].cap += d;
BE2                 return d;
18D             }
57D         }
E13     }
BB3     return 0;
182 }

903 T flow(int s, int t) {
19A     T res = 0;
8CE     while (bfs(s, t)) {
736         fill(iter.begin(), iter.end(), 0);
A96         T d;
AE9         while ((d = dfs(s, t, numeric_limits<T>::max())) > 0)
337             res += d;
91B     }
B50     return res;
CE8 }

8B0 vector<bool> min_cut(int s) {
F71     vector<bool> vis(N, false);
26A     queue<int> q;
451     vis[s] = true;
08B     q.push(s);
14D     while (!q.empty()) {
B1E         int v = q.front();
833         q.pop();
24E         for (auto &e : g[v])
F04             if (e.cap > 0 && !vis[e.to]) {
1E8                 vis[e.to] = true;
6F4                 q.push(e.to);
A56             }
607     }
C81     return vis;
71F }
993 };

```

Min Cost Max Flow

```

/* Min-Cost Max-Flow (SPFA/Bellman-Ford based)
* Complexidade: O(VE·flow) no pior caso.

```

```

* Memoria: O(V + E)
*
* Metodos:
* add_edge(u, v, cap, cost) → aresta u→v com cap e custo
* add_edge(u, v, cap, rcap, cost) → capacidade reversa rcap
* flow(s, t) → {fluxo maximo, custo minimo} de s a t
* flow(s, t, limit) → limita o fluxo total a limit
*
* Observacoes:
* - Suporta custos negativos (usa SPFA em vez de Dijkstra).
* - Sem custos negativos, Dijkstra+potenciais da O(E log V)/aug.
* - Retorna o menor custo para o fluxo maximo obtido.
*/

39C template <typename Cap, typename Cost>
6F3 struct MCMF {
E9B     struct Edge {
C63         int to, rev;
2E2         Cap cap;
CB9         Cost cost;
BFF     };
060 int N;
ED3 vector<vector<Edge>> g;

7E6 MCMF(int n) : N(n), g(n) {}

251 void add_edge(int u, int v, Cap cap, Cost cost, Cap rcap = 0) {
C5D     g[u].push_back({v, (int)g[v].size(), cap, cost});
D90     g[v].push_back({u, (int)g[u].size() - 1, rcap, -cost});
200 }

pair<Cap, Cost> flow(int s, int t,
                    Cap limit = numeric_limits<Cap>::max()) {
C97     pair<Cap, Cost> dist(N, numeric_limits<Cost>::max());
ABA     Cap total_flow = 0;
8AC     Cost total_cost = 0;
8F8

    while (limit > 0) {
DD0         vector<Cost> dist(N, numeric_limits<Cost>::max());
715         vector<bool> in_queue(N, false);
7A2         vector<int> prev_node(N, -1), prev_edge(N, -1);
FF9

        dist[s] = 0;
A93         queue<int> q;
26A         q.push(s);
08B         in_queue[s] = true;
6FD

        while (!q.empty()) {
14D             int v = q.front();
B1E             q.pop();
833             in_queue[v] = false;
F19             for (int i = 0; i < (int)g[v].size(); i++) {
775                 auto &e = g[v][i];
027                 if (e.cap > 0 && dist[v] + e.cost < dist[e.to]) {
AB4                     dist[e.to] = dist[v] + e.cost;
9B1                     prev_node[e.to] = v;
3C0                     prev_edge[e.to] = i;
2E2                     if (!in_queue[e.to]) {
5CA                         in_queue[e.to] = true;
304                         q.push(e.to);
6F4                     }
8A6                 }
5E7             }
168         }
A8B     }

472 if (dist[t] == numeric_limits<Cost>::max()) break;

B96 Cap pushed = limit;
8CE for (int v = t; v != s; v = prev_node[v])
31C     pushed = min(pushed, g[prev_node[v]][prev_edge[v]].cap);

879 for (int v = t; v != s; v = prev_node[v]) {
49B     auto &e = g[prev_node[v]][prev_edge[v]];
A04     e.cap -= pushed;
B9A     g[v][e.rev].cap += pushed;
F38 }

3BD total_flow += pushed;
6CE total_cost += pushed * dist[t];
3FE limit -= pushed;
B73 }

```

```

570     return {total_flow, total_cost};
2AA }
6F3 };

```

Bipartite Matching

```

0AA #include "dinic.hpp"

/* Bipartite Matching via Dinic
* Matching maximo em grafo bipartido.
* Complexidade: O(E * sqrt(V))
*
* Vertices esquerda: [0, n), direita: [n, n+m).
*
* Campos publicos apos match():
* size = tamanho do matching maximo
* match_l[i] = par de i na direita; -1 se livre
* match_r[j] = par de j na esquerda; -1 se livre
*
* Metodos:
* add_edge(i, j) → aresta entre esquerda i e direita j
* match() → executa o matching, retorna tamanho
*/

878 struct BipartiteMatching {
14E     int n, m;
690     Dinic<int> g;
3AE     int S, T;
2AF     vector<int> match_l, match_r;
689     int size = 0;

1FD     BipartiteMatching(int n, int m)
DD3         : n(n), m(m), g(n + m + 2), S(n + m), T(n + m + 1),
779         match_l(n, -1), match_r(m, -1) {
103         for (int i = 0; i < n; i++) g.add_edge(S, i, 1);
CC1         for (int j = 0; j < m; j++) g.add_edge(n + j, T, 1);
B60     }

EE4     void add_edge(int i, int j) { g.add_edge(i, n + j, 1); }

274     int match() {
5D7         size = g.flow(S, T);
830         for (int i = 0; i < n; i++)
B83             for (auto &e : g.g[i])
772                 if (e.to != S && e.cap == 0) {
846                     match_l[i] = e.to - n;
78E                     match_r[e.to - n] = i;
FE7                 }
1C6         return size;
B47     }
CFA };

```

Math

Mint

```

/* mint – Inteiro modular
* Complexidade: O(1) por operacao, O(log mod) para inv()
*
* template<typename T, T mod>
*
* Uso comum:
* using mint = Mint<int, 998244353>;
* using mll = Mint<ll, 998244353LL>;
*
* Operacoes: +, -, *, /, ==, !=, <<
* inv() → inverso modular (mod deve ser primo)
*/

2DB template <typename T, T mod>
4D0 struct Mint {
CA0     T v;
926     Mint(T v = 0) : v(((v % mod) + mod) % mod) {}

EBE     Mint operator+(Mint o) const {
07D         return v + o.v >= mod ? v + o.v - mod : v + o.v;
E93     }

```

```

774 Mint operator-(Mint o) const {
D65     return v - o.v < 0 ? v - o.v + mod : v - o.v;
66C }
3B1 Mint operator*(Mint o) const { return (ll)v * o.v % mod; }
5B6 Mint operator/(Mint o) const { return *this * o.inv(); }

96C Mint &operator+=(Mint o) { return *this = *this + o; }
BB0 Mint &operator-=(Mint o) { return *this = *this - o; }
926 Mint &operator*=(Mint o) { return *this = *this * o; }
E1C Mint &operator/=(Mint o) { return *this = *this / o; }

8D1 bool operator==(Mint o) const { return v == o.v; }
587 bool operator!=(Mint o) const { return v != o.v; }

A45 explicit operator T() const { return v; }

EC5 Mint inv() const {
F05     T a = v, b = mod, x = 1, y = 0;
1B4     while (b) {
295         T q = a / b;
617         a -= q * b;
257         swap(a, b);
67E         x -= q * y;
9DD         swap(x, y);
367     }
EA5     return x;
0CF }

617 friend ostream &operator<<(ostream &os, Mint m) {
B1C     return os << m.v;
09A }
AAA };

```

Fast Exponentiation

```

/* fexp - Exponenciacao rapida modular
* Complexidade: O(log b)
*
* fexp(a, b, mod) → a^b % mod
* modinv(a, mod) → a^{-1} % mod; requer mod primo (Fermat)
*
* Usa __int128 no produto: seguro para mod ate ~9.2·10^{18}.
*/

42B ll fexp(ll a, ll b, ll mod) {
67C     a %= mod;
3F8     if (a < 0) a += mod;
CE0     ll res = 1;
C71     for (; b > 0; b >>= 1, a = (__int128)a * a % mod)
D47         if (b & 1) res = (__int128)res * a % mod;
B50     return res;
A29 }

9E3 ll modinv(ll a, ll mod) {
E74     return fexp(a, mod - 2, mod);
64B }

```

Sieve

```

/* Crivo de Eratostenes
* Complexidade: O(N log log N)
*
* sieve(n) → vetor com todos os primos ate n (inclusive).
*/

705 vector<int> sieve(int n) {
EFA     vector<bool> is_prime(n + 1, true);
FD3     vector<int> primes;
19E     is_prime[0] = is_prime[1] = false;
8A9     for (int p = 2; p <= n; p++) {
29C         if (is_prime[p]) {
BCE             primes.push_back(p);
FAB             if ((ll)p * p <= n)
87F                 for (int i = p * p; i <= n; i += p) is_prime[i] = false;
089         }
284     }
A20     return primes;
912 }

/* Crivo de menor fator primo (SPF) - O(N log log N) build

```

```

* spf[i] = menor primo de i; spf[i] == i indica primo.
* Permite fatorar qualquer x <= n em O(log x).
*
* auto spf = spf_sieve(n);
* factor(x, spf) → pares {primo, expoente} ordenados.
*/

830 vector<int> spf_sieve(int n) {
78F     vector<int> spf(n + 1);
2B8     iota(spf.begin(), spf.end(), 0);
225     for (int i = 2; (ll)i * i <= n; i++)
DD0         if (spf[i] == i)
C0A             for (int j = i * i; j <= n; j += i)
480                 if (spf[j] == j) spf[j] = i;
D4F     return spf;
2DC }

4FF vector<pair<int, int>> factor(int x, const vector<int> &spf) {
490     vector<pair<int, int>> f;
40E     while (x > 1) {
239         int p = spf[x], e = 0;
FA7         while (x % p == 0) {
43F             x /= p;
1A3             e++;
E87         }
6BF         f.push_back({p, e});
760     }
ABE     return f;
B2C }

/* Divisores de x fora de ordem - O(d(x)).
* Requer x dentro do crivo usado para gerar spf.
* Aplique sort() ao retorno quando precisar de ordem.
*/

17D vector<int> divisors(int x, const vector<int> &spf) {
FC1     vector<int> divs = {1};
D07     for (auto [p, e] : factor(x, spf)) {
F49         int sz = divs.size(), pk = 1;
6E5         for (int k = 0; k < e; k++) {
DC5             pk *= p;
BA1             for (int i = 0; i < sz; i++) divs.push_back(divs[i] * pk);
FB5         }
261     }
81C     return divs;
C23 }

Euler Phi

/* Crivo Linear - O(N)
*
* linear_sieve(n) → {primes, phi}
* primes : todos os primos ate n (inclusive)
* phi    : phi[i] = φ(i) para i em [0, n]
*/

9A7 pair<vector<int>, vector<int>> linear_sieve(int n) {
303     vector<int> phi(n + 1, 0), primes;
614     vector<bool> composite(n + 1, false);
678     phi[1] = 1;
508     for (int i = 2; i <= n; i++) {
C54         if (!composite[i]) {
E74             primes.push_back(i);
ABD             phi[i] = i - 1;
6DB         }
3B9         for (int p : primes) {
A22             if ((ll)i * p > n) break;
847             composite[i * p] = true;
569             if (i % p == 0) {
8EC                 phi[i * p] = phi[i] * p;
C2B                 break;
792             } else {
812                 phi[i * p] = phi[i] * (p - 1);
988             }
C4A         }
839     }
6EF     return {primes, phi};

```

```

415 }

Number Theory

/* GCD / LCM - O(log min(a, b)) */
9C4 ll gcd(ll a, ll b) {
D0B     return b ? gcd(b, a % b) : a;
225 }
026 ll lcm(ll a, ll b) {
8B8     return a / gcd(a, b) * b;
15C }

/* Divisores de n - O(sqrt(N))
* Retorna lista de divisores em ordem crescente.
*/
5FA vector<ll> divisors(ll n) {
17E     vector<ll> lo, hi;
148     for (ll i = 1; i * i <= n; i++) {
C06         if (n % i == 0) {
1EB             lo.push_back(i);
9C3             if (i != n / i) hi.push_back(n / i);
B5B         }
2DE     }
FC1     reverse(hi.begin(), hi.end());
3AF     lo.insert(lo.end(), hi.begin(), hi.end());
253     return lo;
B9C }

/* Fatoracao em primos - O(sqrt(N))
* Retorna lista de {primo, expoente} em ordem crescente.
*/
7E3 vector<pair<ll, int>> factorize(ll n) {
618     vector<pair<ll, int>> factors;
D2E     for (ll p = 2; p * p <= n; p++) {
80A         if (n % p == 0) {
A01             int exp = 0;
03E             while (n % p == 0) {
F4A                 n /= p;
666                 exp++;
6C9             }
13A             factors.push_back({p, exp});
2E9         }
47D     }
992     if (n > 1) factors.push_back({n, 1});
FA9     return factors;
92A }

/* CRT - Teorema Chines dos Restos - O(log min(m1, m2))
* Retorna {x, lcm} com x ≡ a1 (mod m1), x ≡ a2 (mod m2).
* Retorna {0, 0} se o sistema for incompativel.
*/
8A9 pair<ll, ll> crt(ll a1, ll m1, ll a2, ll m2) {
8F7     ll g = gcd(m1, m2);
916     if ((a2 - a1) % g != 0) return {0, 0};
931     ll mod = m1 / g * m2;
D0B     ll a = m1 / g, b = m2 / g, x = 1, y = 0;
E58     for (ll ta = a, tb = b; tb;) {
B40         ll q = ta / tb;
4AA         ta -= q * tb;
153         swap(ta, tb);
67E         x -= q * y;
9DD         swap(x, y);
54E     }
581     x = (x % (m2 / g) + m2 / g) % (m2 / g);
A62     ll step = (a2 - a1) / g % (m2 / g) * x % (m2 / g);
5FD     ll result = (a1 + m1 * step) % mod;
1EB     return {(result + mod) % mod, mod};
529 }

/* Fatoracao com primos pre-computados - O(π(sqrt(N)))
* Util apos sieve(sqrt(n)) para fatorar varios numeros.
* primos deve conter todos os primos ate sqrt(n).
*/
165 vector<pair<ll, int>> factorize(ll n, const vector<int> &pr) {
618     vector<pair<ll, int>> factors;
38E     for (int p : pr) {

```

```

851     if ((ll)p * p > n) break;
80A     if (n % p == 0) {
A01         int exp = 0;
03E         while (n % p == 0) {
F4A             n /= p;
666             exp++;
6C9         }
13A         factors.push_back({p, exp});
2E9     }
6EA }
992 if (n > 1) factors.push_back({n, 1});
FA9 return factors;
927 }

```

Miller Rabin

```

/* Miller-Rabin + Pollard Rho
*
* is_prime(n) → 0(log² n), determinístico para n < 3.3·10¹⁸
* factorize(n) → 0(n^(1/4) log n) esperado; repete fatores
*
* Para fatores distintos, use sort + unique.
*/

DA8 ll mulmod(ll a, ll b, ll m) {
C2A     return (__int128)a * b % m;
7D4 }

0A0 bool is_prime(ll n) {
5A7     if (n < 2) return false;
0A8     if (n == 2 || n == 3 || n == 5 || n == 7) return true;
3CE     if (n % 2 == 0 || n % 3 == 0) return false;
AC7     ll d = n - 1;
898     int r = 0;
5B7     while (d % 2 == 0) d /= 2, r++;
C12     for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
89B         if (a >= n) continue;
8C8         ll x = 1, base = a % n, exp = d;
006         for (; exp > 0; exp >>= 1, base = mulmod(base, base, n))
5E6             if (exp & 1) x = mulmod(x, base, n);
8A3         if (x == 1 || x == n - 1) continue;
30D         bool composite = true;
21B         for (int i = 0; i < r - 1; i++) {
F38             x = mulmod(x, x, n);
A11             if (x == n - 1) {
827                 composite = false;
C2B                 break;
789             }
4C7         }
4F2         if (composite) return false;
C63     }
8A6     return true;
462 }

711 ll pollard_rho(ll n) {
55A     if (n % 2 == 0) return 2;
ADB     ll x = rand() % (n - 2) + 2, y = x, d = 1;
A4B     ll c = rand() % (n - 1) + 1;
4EB     while (d == 1) {
C17         x = (mulmod(x, x, n) + c) % n;
4EE         y = (mulmod(y, y, n) + c) % n;
4EE         y = (mulmod(y, y, n) + c) % n;
DD8         d = __gcd(abs(x - y), n);
565     }
9EC     return d == n ? pollard_rho(n) : d;
C00 }

6C2 vector<ll> factorize(ll n) {
1B9     if (n == 1) return {};
970     if (is_prime(n)) return {n};
B32     ll d = pollard_rho(n);
8AB     auto l = factorize(d), r = factorize(n / d);
3AF     l.insert(l.end(), r.begin(), r.end());
792     return l;
928 }

```

Matrix

```

/* Matrix – Multiplicacao e exponenciacao de matriz quadrada
* Complexidade: O(n³ log k) para M^k
*
* Use com mint/mlt para operacoes modulares automaticas.
*
* Matrix<T> M(n) → identidade nxn
* Matrix<T> M(n,0) → matriz nxn zerada
* M[i][j] → acesso direto
* A * B → multiplicacao O(n³)
* M ^ k → M^k por exponenciacao rapida
*/

67A template <typename T>
BF3 struct Matrix {
1A8     int n;
C24     vector<vector<T>> mat;

049     Matrix(int n, int identity = 1)
9AE         : n(n), mat(n, vector<T>(n, T(0))) {
340         if (identity)
3BD             for (int i = 0; i < n; i++) mat[i][i] = T(1);
6ED     }

90D     vector<T> &operator[](int i) { return mat[i]; }
699     const vector<T> &operator[](int i) const { return mat[i]; }

AFB     Matrix operator*(const Matrix &rhs) const {
83B         Matrix res(n, 0);
830         for (int i = 0; i < n; i++)
E22             for (int k = 0; k < n; k++)
E64                 if (mat[i][k] != T(0))
F90                     for (int j = 0; j < n; j++)
C55                         res[i][j] = res[i][j] + mat[i][k] * rhs.mat[k][j];
B50         return res;
893     }

348     Matrix operator^(ll b) const {
2DF         Matrix res(n), a = *this;
0EB         for (; b > 0; b >>= 1, a = a * a)
FC2             if (b & 1) res = res * a;
B50         return res;
1B3     }
ACF };

```

Gaussian Elimination

```

/* Eliminacao Gaussiana (RREF) – O(N² * M)
* Funciona com double ou Mint<T, mod>.
*
* gauss(A, b) → resolve Ax = b.
* Retorna {x, true} se unica; false nos demais casos.
*/

67A template <typename T>
5D7 pair<vector<T>, bool> gauss(vector<vector<T>> A, vector<T> b) {
FE3     int n = A.size();
3E0     for (int i = 0; i < n; i++) A[i].push_back(b[i]);
3D7     int m = n + 1, rank = 0;
079     for (int col = 0; col < n && rank < n; col++) {
9D3         int sel = rank;
F97         for (int i = rank + 1; i < n; i++)
E75             if (abs(A[i][col]) > abs(A[sel][col])) sel = i;
9E6         if (!(bool)A[sel][col]) continue;
D8D         swap(A[sel], A[rank]);
B46         T iv = T(1) / A[rank][col];
C24         for (int j = col; j < m; j++) A[rank][j] = A[rank][j] * iv;
603         for (int i = 0; i < n; i++) {
BC7             if (i == rank || !(bool)A[i][col]) continue;
BEF             T f = A[i][col];
69A             for (int j = col; j < m; j++)
E47                 A[i][j] = A[i][j] - f * A[rank][j];
371         }
56E         rank++;
3F2     }
175     if (rank < n) return {{}, false};
A1B     vector<T> x(n);

```

```

B95     for (int i = 0; i < n; i++) x[i] = A[i].back();
079     return {x, true};
724 }

```

Linear Basis

```

/* Linear Basis (Base Linear XOR)
* Complexidade: O(B) por operacao; B = numero de bits.
* Memoria: O(B)
*
* Metodos:
* insert(x) → true se x e linearmente independente
* contains(x) → x pertence ao espaco gerado pela base
* max_xor(x=0) → maior x XOR subconjunto da base
* min_xor() → menor XOR nao-nulo; 0 se base incompleta
* size() → numero de elementos linearmente independentes
* merge(other) → insere os elementos de other
*
* Observacoes:
* - Na forma reduzida, cada pivo aparece em uma so linha.
* - Para o k-esimo XOR, chame reduce() antes de kth().
*/

538 template <typename T = ll, int B = 63>
F45 struct LinearBasis {
E9C     array<T, B + 1> basis{};
1FC     int sz = 0;

220     bool insert(T x) {
C8C         for (int i = B; i >= 0; i--) {
760             if (!(x >> i) & 1) continue;
617             if (!basis[i]) {
C6C                 basis[i] = x;
EB9                 sz++;
8A6                 return true;
50C             }
6B0             x ^= basis[i];
8A2         }
D1F         return false;
CD7     }

394     bool contains(T x) const {
C8C         for (int i = B; i >= 0; i--) {
760             if (!(x >> i) & 1) continue;
260             if (!basis[i]) return false;
6B0             x ^= basis[i];
1A9         }
8A6         return true;
189     }

C3B     T max_xor(T x = 0) const {
3CC         for (int i = B; i >= 0; i--) x = max(x, x ^ basis[i]);
EA5         return x;
74B     }

BBB     T min_xor() const {
F56         for (int i = 0; i <= B; i++)
15D             if (basis[i]) return basis[i];
BB3         return 0;
5FC     }

21A     int size() const { return sz; }

AC0     void merge(const LinearBasis &other) {
086         for (int i = B; i >= 0; i--)
D67             if (other.basis[i]) insert(other.basis[i]);
1CA     }

/* Reduz a base: cada pivo aparece em uma unica linha.
* Necessario antes de kth(); gera os 2^sz XORs distintos.
*/

93A     void reduce() {
C8C         for (int i = B; i >= 0; i--) {
F14             if (!basis[i]) continue;
E36             for (int j = i + 1; j <= B; j++)
7E7                 if ((basis[j] >> i) & 1) basis[j] ^= basis[i];
CA5         }
563     }

```

```

/* k-esimo menor XOR, 0-indexado, entre 2^sz subconjuntos.
* Requer reduce(). k=0 e 0; k=1 e o menor nao-nulo.
*/
0B4 T kth(ll k) const {
E55   vector<T> elems;
F56   for (int i = 0; i <= B; i++)
210     if (basis[i]) elems.push_back(basis[i]);
    // Ordem crescente do bit mais significativo.
FAC   if (k >= (1LL << (int)elems.size()))
DAA     return -1; // k fora do range
19A   T res = 0;
687   for (int i = 0; i < (int)elems.size(); i++)
A7C     if ((k >> i) & 1) res ^= elems[i];
B50   return res;
B1A }
DE8 };

```

Unimodal Search

```

/* Busca unimodal – f deve ser unimodal no intervalo dado.
*
* ternary_min/max(lo, hi, f) → inteiros, O(log N)
* golden_min/max(lo, hi, f, iters=200) → contínuo, O(iters)
*/
// Ternary search para mínimo em inteiros [lo, hi]
398 template <typename F>
7B4 ll ternary_min(ll lo, ll hi, F f) {
960   while (hi - lo > 2) {
27E     ll m1 = lo + (hi - lo) / 3;
D61     ll m2 = hi - (hi - lo) / 3;
2EE     if (f(m1) <= f(m2)) hi = m2;
F67     else lo = m1;
794   }
23F   ll best = lo;
AC1   for (ll x = lo + 1; x <= hi; x++)
EB8     if (f(x) < f(best)) best = x;
F26   return best;
67D }

// Ternary search para máximo em inteiros [lo, hi]
398 template <typename F>
810 ll ternary_max(ll lo, ll hi, F f) {
960   while (hi - lo > 2) {
27E     ll m1 = lo + (hi - lo) / 3;
D61     ll m2 = hi - (hi - lo) / 3;
750     if (f(m1) >= f(m2)) hi = m2;
F67     else lo = m1;
F46   }
23F   ll best = lo;
AC1   for (ll x = lo + 1; x <= hi; x++)
592     if (f(x) > f(best)) best = x;
F26   return best;
1C7 }

```

```

// Golden section search para mínimo em contínuo [lo, hi]
398 template <typename F>
938 double golden_min(double lo, double hi, F f, int iters = 200) {
84E   const double phi = (sqrt(5.0) - 1.0) / 2.0; // ≈ 0.618
0DD   double m1 = hi - phi * (hi - lo);
86C   double m2 = lo + phi * (hi - lo);
87F   double f1 = f(m1), f2 = f(m2);
5B6   for (int i = 0; i < iters; i++) {
F4D     if (f1 < f2) {
82B       hi = m2;
3E8       m2 = m1;
068       f2 = f1;
3D2       m1 = hi - phi * (hi - lo);
8B5       f1 = f(m1);
735     } else {
CEC       lo = m1;
0DE       m1 = m2;
36A       f1 = f2;
C3A       m2 = lo + phi * (hi - lo);
139       f2 = f(m2);
E4A     }

```

```

DC3   }
E67   return (lo + hi) / 2.0;
2D2 }

// Golden section search para máximo em contínuo [lo, hi]
398 template <typename F>
CEA double golden_max(double lo, double hi, F f, int iters = 200) {
564   return golden_min(
5F7     lo, hi, [&](double x) { return -f(x); }, iters);
2E7 }

```

FFT

```

316 using CD = complex<ld>;

/* FFT – Fast Fourier Transform
* Complexidade: O(N log N)
*
* fft(a) → transforma a no lugar; |a| e potencia de 2
* conv(a, b) → convolucao: c[k] = Σ a[i]*b[k-i]
*
* Precisão: seguro se (Σa²+Σb²)*log₂N < 9·10¹⁴.
* Para coeficientes grandes ou módulo, use fft-mod ou NTT.
*/
B4C void fft(vector<CD> &a) {
820   int n = a.size(), log_n = 31 - __builtin_clz(n);
F82   static vector<complex<long double>> R(2, 1);
6B4   static vector<CD> rt(2, 1);
AD8   for (static int k = 2; k < n; k *= 2) {
411     auto x = polar(1.0L, acos(-1.0L) / k);
9D9     R.resize(n);
335     rt.resize(n);
1D3     for (int i = k; i < 2 * k; i++)
CD4       rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
040   }

808   vector<int> rev(n);
830   for (int i = 0; i < n; i++)
F9A     rev[i] = (rev[i / 2] | (i & 1) << log_n) / 2;
830   for (int i = 0; i < n; i++)
A75     if (i < rev[i]) swap(a[i], a[rev[i]]);

657   for (int k = 1; k < n; k *= 2)
1E5     for (int i = 0; i < n; i += 2 * k)
0C2       for (int j = 0; j < k; j++) {
CD2         auto x = (ld *)&rt[j + k], y = (ld *)&a[i + j + k];
259         CD z(x[0] * y[0] - x[1] * y[1],
45D           x[0] * y[1] + x[1] * y[0]);
20A         a[i + j + k] = a[i + j] - z;
1B0         a[i + j] += z;
707       }
960 }

```

```

17B vector<ld> conv(const vector<ld> &a, const vector<ld> &b) {
F88   if (a.empty() || b.empty()) return {};
BBB   vector<ld> res(a.size() + b.size() - 1);
E9A   int n = 1 << (32 - __builtin_clz(res.size()));
576   vector<CD> in(n), out(n);
F83   copy(begin(a), end(a), begin(in));
D38   for (int i = 0; i < (int)b.size(); i++) in[i].imag(b[i]);
21A   fft(in);
11C   for (auto &x : in) x *= x;
830   for (int i = 0; i < n; i++)
5DA     out[i] = in[-i & (n - 1)] - conj(in[i]);
3D7   fft(out);
2D6   for (int i = 0; i < (int)res.size(); i++)
A2F     res[i] = imag(out[i]) / (4 * n);
B50   return res;
B6D }

```

FFT Mod

```

240 #include "fft.hpp"

/* FFT Mod – Convolucao modular com mod arbitrario
* Complexidade: O(N log N) (~2x mais lento que FFT ou NTT)
*

```

```

* convMod<mod>(a, b) → c[k] = (Σ a[i]*b[k-i]) % mod
*
* Entradas devem estar em [0, mod).
* Seguro se N·log₂N·mod < 8.6·10¹⁴.
*
* Sem mod: use cut = 1<<15 e remova os operadores % mod.
*/

```

```

3F9 template <int mod>
2A7 vector<ll> convMod(const vector<ll> &a, const vector<ll> &b) {
F88   if (a.empty() || b.empty()) return {};
290   vector<ll> res(a.size() + b.size() - 1);
07A   int log_n = 32 - __builtin_clz(res.size());
8E1   int n = 1 << log_n;
    // Para alta precisao, use 1<<15 e tire os % mod abaixo.
528   int cut = (int)sqrt(mod);
584   vector<CD> L(n), R(n), outs(n), outl(n);
5B0   for (int i = 0; i < (int)a.size(); i++)
8B7     L[i] = CD((int)a[i] / cut, (int)a[i] % cut);
81B   for (int i = 0; i < (int)b.size(); i++)
F55     R[i] = CD((int)b[i] / cut, (int)b[i] % cut);
5D5   fft(L), fft(R);
603   for (int i = 0; i < n; i++) {
39D     int j = -i & (n - 1);
65E     outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
482     outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / CD(0, 1);
1C6   }
D08   fft(outl), fft(outs);
6D1   for (int i = 0; i < (int)res.size(); i++) {
    // alta precisao: tire os % mod nas linhas abaixo
54F     ll av = (ll)(real(outl[i]) + .5) % mod;
FA2     ll bv = (ll)(imag(outl[i]) + .5) + (ll)(real(outs[i]) + .5);
A36     ll cv = (ll)(imag(outs[i]) + .5);
557     res[i] = ((av * cut + bv) % mod * cut + cv) % mod;
F04   }
B50   return res;
CA2 }

```

NTT

```

/* NTT – Number Theoretic Transform
* Complexidade: O(N log N)
*
* NTT<mod, g>::conv(a, b) → c[k] = (Σ a[i]*b[k-i]) % mod
*
* Entradas devem estar em [0, mod).
* mod e primo 2^a·b+1; g e sua raiz primitiva.
*
* Opcoes de {mod, g}:
* {998244353, 3} → 119·2²³+1, mais comum, N ≤ 2²³
* {469762049, 3} → 7·2²⁶+1, N ≤ 2²⁶
* {167772161, 3} → 5·2²⁵+1, N ≤ 2²⁵
* {2013265921, 31} → 15·2²⁷+1, N ≤ 2²⁷ (mod > 10⁹)
* {2281701377, 3} → 17·2²⁷+1, N ≤ 2²⁷ (mod > 10⁹)
* {7340033, 5} → 7·2²⁰+1, N ≤ 2²⁰ (menor mod)
* {9223372036737335297LL, 3} → 549755813881·2²⁴+1, N ≤ 2²⁴
* (mod-2⁶³, ll)
*/

```

```

E34 template <ll mod, ll g>
F12 struct NTT {
D3B   static ll fexp(ll a, ll b) {
CE0     ll res = 1;
67C     a %= mod;
FF0     for (; b > 0; b >= 1, a = a * a % mod)
602       if (b & 1) res = res * a % mod;
B50     return res;
B0B   }

936   static void ntt(vector<ll> &a) {
820     int n = a.size(), log_n = 31 - __builtin_clz(n);
D51     static vector<ll> rt(2, 1);
8EE     for (static int k = 2, s = 2; k < n; k *= 2, s++) {
335       rt.resize(n);
277       ll z[] = {1, fexp(g, mod >> s)};
1D3       for (int i = k; i < 2 * k; i++)

```

```

256     rt[i] = rt[i / 2] * z[i & 1] % mod;
D00 }
808 vector<int> rev(n);
830 for (int i = 0; i < n; i++)
F9A     rev[i] = (rev[i / 2] | (i & 1) << log_n) / 2;
830 for (int i = 0; i < n; i++)
A75     if (i < rev[i]) swap(a[i], a[rev[i]]);
657 for (int k = 1; k < n; k *= 2)
1E5     for (int i = 0; i < n; i += 2 * k)
0C2         for (int j = 0; j < k; j++) {
86E             ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
598             a[i + j + k] = ai - z + (z > ai ? mod : 0);
4B8             ai += z - (ai + z >= mod ? mod : 0);
D6A         }
82E     }

46D static vector<ll> conv(vector<ll> a, vector<ll> b) {
F88     if (a.empty() || b.empty()) return {};
5C2     int s = a.size() + b.size() - 1;
DE2     int n = 1 << (32 - __builtin_clz(s));
667     a.resize(n), b.resize(n); ntt(a), ntt(b);
DDC     ll inv = fexp(n, mod - 2);
6F8     vector<ll> out(n);
830     for (int i = 0; i < n; i++)
50E         out[-i & (n - 1)] = a[i] * b[i] % mod * inv % mod;
EC9     ntt(out);
C20     return {out.begin(), out.begin() + s};
659 }
F65 };

```

FWHT

```

/* FWHT – Fast Walsh-Hadamard Transform
 * Complexidade: O(N log N)
 *
 * Convolucao bitwise: c[k] = Σ a[i]*b[j] onde i op j = k
 *
 * fwht_conv<'^'>(a, b) → convolucao XOR
 * fwht_conv<'|'>(a, b) → convolucao OR
 * fwht_conv<'&'>(a, b) → convolucao AND
 *
 * Ajusta os tamanhos para a proxima potencia de 2.
 */

597 template <char op>
823 void fwht(vector<ll> &a, bool inv = false) {
94D     int n = a.size();
980     for (int len = 1; len < n; len *= 2)
EBC         for (int i = 0; i < n; i += 2 * len)
7AB             for (int j = 0; j < len; j++) {
032                 ll u = a[i + j], v = a[i + j + len];
FBD                 if (op == '^') a[i + j] = u + v, a[i + j + len] = u - v;
02F                 if (op == '|') a[i + j + len] = v + (inv ? -u : +u);
729                 if (op == '&') a[i + j] = u + (inv ? -v : +v);
1C4             }
609             if (op == '^' && inv)
F33                 for (auto &x : a) x /= n;
643     }

597 template <char op>
E0F vector<ll> fwht_conv(vector<ll> a, vector<ll> b) {
43E     int n = 1;
3B9     while (n < (int)max(a.size(), b.size())) n *= 2;
711     a.resize(n);
DD6     b.resize(n);
883     fwht<op>(a);
671     fwht<op>(b);
34E     for (int i = 0; i < n; i++) a[i] *= b[i];
9D7     fwht<op>(a, true);
3F5     return a;
22D }

```

String

KMP

```

/* KMP (Knuth-Morris-Pratt)
 * failure[i] = tamanho do maior prefixo proprio de s[0..i]
 * que e tambem sufixo.
 * Complexidade: O(N) build, O(N + M) busca.
 *
 * Aplicacoes:
 * - Busca de padrao P em texto T em O(|P| + |T|).
 * - Menor periodo da string: N - failure[N-1].
 * - Contar ocorrencias de todos os prefixos como sufixos.
 */

090 vector<int> kmp_failure(const string &s) {
163     int n = s.size();
D11     vector<int> f(n, 0);
6F5     for (int i = 1; i < n; i++) {
844         int j = f[i - 1];
424         while (j > 0 && s[i] != s[j]) j = f[j - 1];
04B         if (s[i] == s[j]) j++;
199         f[i] = j;
D74     }
ABE     return f;
8B7 }

// Índices de todas as ocorrencias de pat em txt.
A0D vector<int> kmp_search(const string &pat, const string &txt) {
921     string s = pat + "$" + txt;
3EC     vector<int> f = kmp_failure(s);
11E     int p = pat.size();
927     vector<int> pos;
D6B     for (int i = p + 1; i < (int)s.size(); i++)
06B         if (f[i] == p) pos.push_back(i - 2 * p);
D75     return pos;
F47 }

```

Z Function

```

/* Z-Function
 * z[i] = maior prefixo de s que coincide com s[i..].
 * z[0] = 0 por convencao.
 * Complexidade: O(N) build.
 *
 * Aplicacoes:
 * - Busca P em T: calcule Z de P + "$" + T.
 * - Ocorrencias de prefixo k: conte z[i] >= k.
 * - Menor periodo: menor p | N com z[p] == N-p.
 */

7AD vector<int> z_function(const string &s) {
163     int n = s.size();
2B1     vector<int> z(n, 0);
A5C     for (int i = 1, l = 0, r = 0; i < n; i++) {
20E         if (i < r) z[i] = min(r - i, z[i - l]);
4ED         while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
B76         if (i + z[i] > r) l = i, r = i + z[i];
3BB     }
070     return z;
197 }

// Índices de todas as ocorrencias de pattern em text.
430 vector<int> z_search(const string &pattern, const string &text) {
365     string s = pattern + "$" + text;
EA6     vector<int> z = z_function(s);
E6D     int p = pattern.size();
FCC     vector<int> positions;
D6B     for (int i = p + 1; i < (int)s.size(); i++)
484         if (z[i] == p) positions.push_back(i - p - 1);
C4C     return positions;
BBF }

```

String Hash

```

/* String Hashing (Single e Double Hash)
 * Hash polinomial: build O(N), comparacao O(1).

```

```

 * Complexidade: O(N) build, O(1) query.
 * Memoria: O(N)
 *
 * SingleHash: um par (MOD, BASE), com baixo risco de colisao.
 * DoubleHash: dois pares independentes.
 *
 * Uso:
 *     SingleHash h(s);
 *     h.get(l, r); // hash de s[l..r]
 *     h.get(l, r) == h.get(l2, r2); // compara substrings
 */

/* --- Single Hash --- */
55A struct SingleHash {
029     static const ll MOD = (1LL << 61) - 1;
56C     static const ll BASE;

F42     vector<ll> h, pw;

30E     static ll mul(ll a, ll b) { return ((__uint128_t)a * b % MOD); }

67F     static ll add(ll a, ll b) {
9F4         a += b;
BF2         return a >= MOD ? a - MOD : a;
4E8     }

82B     SingleHash() {}

107     SingleHash(const string &s)
2F6         : h(s.size() + 1, 0), pw(s.size() + 1, 1) {
C46         for (int i = 0; i < (int)s.size(); i++) {
4DB             h[i + 1] = add(mul(h[i], BASE), s[i]);
F0F             pw[i + 1] = mul(pw[i], BASE);
0FA         }
485     }

BA8     ll get(int l, int r) const {
128         return add(h[r + 1], MOD - mul(h[l], pw[r - l + 1]));
358     }
4EF };

E2C const ll SingleHash::BASE = []() {
862     mt19937_64 rng(
854         chrono::steady_clock::now().time_since_epoch().count());
2E2     return uniform_int_distribution<ll>(1e9,
86D         SingleHash::MOD - 1)(rng);
7E5 }();

/* --- Double Hash --- */
2AF struct DoubleHash {
172     static const ll MOD1 = (1LL << 61) - 1;
D89     static const ll MOD2 = 1e9 + 9;
2BF     static const ll BASE1, BASE2;

23A     vector<ll> h1, h2, pw1, pw2;

780     static ll mull1(ll a, ll b) {
C20         return ((__uint128_t)a * b % MOD1;
583     }

876     static ll add1(ll a, ll b) {
9F4         a += b;
B86         return a >= MOD1 ? a - MOD1 : a;
577     }

997     static ll mul2(ll a, ll b) { return (__int128)a * b % MOD2; }
F18     static ll add2(ll a, ll b) {
9F4         a += b;
156         return a >= MOD2 ? a - MOD2 : a;
AC6     }

A3E     DoubleHash() {}

3C9     DoubleHash(const string &s)
01A         : h1(s.size() + 1, 0), h2(s.size() + 1, 0),
4CF             pw1(s.size() + 1, 1), pw2(s.size() + 1, 1) {
C46         for (int i = 0; i < (int)s.size(); i++) {
9F0             h1[i + 1] = add1(mul1(h1[i], BASE1), s[i]);
128             h2[i + 1] = add2(mul2(h2[i], BASE2), s[i]);
562             pw1[i + 1] = mul1(pw1[i], BASE1);
961             pw2[i + 1] = mul2(pw2[i], BASE2);
706         }
366     }

```

```

// Par de hashes de s[l..r], inclusive.
C89 pair<ll, ll> get(int l, int r) const {
DC7   ll a = add1(h1[r + 1], MOD1 - mul1(h1[l], pw1[r - l + 1]));
932   ll b = add2(h2[r + 1], MOD2 - mul2(h2[l], pw2[r - l + 1]));
594   return {a, b};
731 }
3DF };

BBD const ll DoubleHash::BASE1 = []() {
862   mt19937_64 rng(
854     chrono::steady_clock::now().time_since_epoch().count());
2E2   return uniform_int_distribution<ll>(1e9,
263     DoubleHash::MOD1 - 1)(rng);
C70 }());

0BD const ll DoubleHash::BASE2 = []() {
862   mt19937_64 rng(
F11     chrono::steady_clock::now().time_since_epoch().count() + 1);
2E2   return uniform_int_distribution<ll>(1e9,
91C     DoubleHash::MOD2 - 1)(rng);
13F }());

```

Trie

```

/* Trie (Prefix Tree)
 * Insercao e busca de strings em O(|s|).
 * Memoria: O(N * ALPHA), N = total de caracteres.
 * Alfabeto: [BASE, BASE + ALPHA).
 */
71F template <int ALPHA = 26, char BASE = 'a'>
71A struct Trie {
BF2   struct Node {
C02     int ch[ALPHA];
0F9     int cnt;
302     int end;
D7F     Node() : cnt(0), end(0) { fill(ch, ch + ALPHA, -1); }
B69   };
F5B   vector<Node> t;
8E7   Trie() { t.emplace_back(); }
E7D   void insert(const string &s) {
3F3     int node = 0;
B4F     for (char c : s) {
E9F       int id = c - BASE;
19E       if (t[node].ch[id] == -1) {
013         t[node].ch[id] = t.size();
83D         t.emplace_back();
838       }
729       node = t[node].ch[id];
59E       t[node].cnt++;
304     }
879     t[node].end++;
2C4   }
BA9   bool search(const string &s) const {
3F3     int node = 0;
B4F     for (char c : s) {
E9F       int id = c - BASE;
D92       if (t[node].ch[id] == -1) return false;
729       node = t[node].ch[id];
047     }
395     return t[node].end > 0;
3FB   }
0B1   int count_prefix(const string &s) const {
3F3     int node = 0;
B4F     for (char c : s) {
E9F       int id = c - BASE;
2AA       if (t[node].ch[id] == -1) return 0;
729       node = t[node].ch[id];
88D     }
D17     return t[node].cnt;
673   }
4E2   int count(const string &s) const {
3F3     int node = 0;

```

```

B4F   for (char c : s) {
E9F     int id = c - BASE;
2AA     if (t[node].ch[id] == -1) return 0;
729     node = t[node].ch[id];
88D   }
D51   return t[node].end;
899 }
25C };

```

Manacher

```

/* Manacher's Algorithm (Palindrome Detection)
 * Encontra todos os palindromos em tempo linear.
 * Complexidade: O(N) onde N e o tamanho da string.
 * Memoria: O(N)
 * Funcoes:
 * - manacher(s): maior palindromo centrado em i.
 * - palindrome: testa s[i..j] em O(1).
 * - pal_begin(s): maior palindromo comecando em i.
 * - pal_end(s): maior palindromo terminando em i.
 */
67A template <typename T>
44D vector<int> manacher(const T &s) {
18F   int l = 0, r = -1, n = s.size();
FC9   vector<int> d1(n), d2(n);
603   for (int i = 0; i < n; i++) {
821     int k = i > r ? 1 : min(d1[l + r - i], r - i);
61A     while (i + k < n && i - k >= 0 && s[i + k] == s[i - k]) k++;
61E     d1[i] = k--;
9F6     if (i + k > r) l = i - k, r = i + k;
950   }
E03   l = 0, r = -1;
603   for (int i = 0; i < n; i++) {
1A6     int k = i > r ? 0 : min(d2[l + r - i + 1], r - i + 1);
AC1     k++;
A94     while (i + k <= n && i - k >= 0 && s[i + k - 1] == s[i - k])
AC1       k++;
EAA     d2[i] = --k;
26D     if (i + k - 1 > r) l = i - k, r = i + k - 1;
4FE   }
1B2   vector<int> res(2 * n - 1);
5CB   for (int i = 0; i < n; i++) res[2 * i] = 2 * d1[i] - 1;
891   for (int i = 0; i < n - 1; i++)
748     res[2 * i + 1] = 2 * d2[i + 1];
B50   return res;
85A }

67A template <typename T>
BF0 struct palindrome {
F97   vector<int> man;
B2D   palindrome(const T &s) : man(manacher(s)) {}
1E7   bool query(int i, int j) { return man[i + j] >= j - i + 1; }
933 };

67A template <typename T>
10D vector<int> pal_begin(const T &s) {
542   int n = s.size() - 1;
A23   vector<int> res(s.size());
FDE   palindrome<T> p(s);
371   res[n] = 1;
45B   for (int i = n - 1; i >= 0; i--) {
D9A     res[i] = min(res[i + 1] + 2, n - i + 1);
F2F     while (!p.query(i, i + res[i] - 1)) res[i]--;
DD9   }
B50   return res;
212 }

67A template <typename T>
934 vector<int> pal_end(const T &s) {
A23   vector<int> res(s.size());
FDE   palindrome<T> p(s);
4EC   res[0] = 1;
88E   for (int i = 1; i < s.size(); i++) {
61D     res[i] = min(res[i - 1] + 2, i + 1);
488     while (!p.query(i - res[i] + 1, i)) res[i]--;

```

```

5A2   }
B50   return res;
A2C }

```

Suffix Array (NlogN)

```

/* Suffix Array O(N log N) - Prefix Doubling com Count Sort
 * sa[i] = indice do i-esimo sufixo em ordem lexicografica.
 * rk[i] = rank do sufixo s[i..] (inverso de sa).
 * lcp[i] = LCP entre sa[i-1] e sa[i] (lcp[0] = 0).
 * Adiciona '$' ao final da string internamente.
 * Complexidade: O(N log N) build, O(|P| log N) search.
 *
 * Aplicacoes:
 * - Busca de padrao P: search(p) retorna [lo, hi) no SA.
 * - Substring mais longa repetida: *max_element(lcp).
 * - Numero de substrings distintas: N*(N+1)/2 - sum(lcp).
 * - LCS de duas strings: SuffixArray::lcs(s, t).
 */
3F4 struct SuffixArray {
AC0   string s;
58B   vector<int> sa, rk, lcp;
264   SuffixArray() {}
357   SuffixArray(string s) : s(s) {
6F2     build();
E8F     build_lcp();
A87   }
0A8   void build() {
FC8     s += '$';
163     int n = s.size();
FB9     sa.resize(n);
9BE     rk.resize(n);
413     vector<pair<char, int>> a(n);
490     for (int i = 0; i < n; i++) a[i] = {s[i], i};
3F8     sort(a.begin(), a.end());
67D     for (int i = 0; i < n; i++) sa[i] = a[i].second;
F27     rk[sa[0]] = 0;
6F5     for (int i = 1; i < n; i++) {
7BD       int inc = a[i].first != a[i - 1].first;
6C1       rk[sa[i]] = rk[sa[i - 1]] + inc;
84B     }
8A4     for (int k = 0; (1 << k) < n; k++) {
830       for (int i = 0; i < n; i++)
D82         sa[i] = (sa[i] - (1 << k) + n) % n;
188       count_sort();
037       vector<int> new_rk(n);
CCE       new_rk[sa[0]] = 0;
C6D       auto snd = [&](int x) { return rk[(x + (1 << k)) % n]; };
6F5       for (int i = 1; i < n; i++) {
CD1         pair<int, int> prev = {rk[sa[i - 1]], snd(sa[i - 1])};
3F0         pair<int, int> now = {rk[sa[i]], snd(sa[i])};
AAD         new_rk[sa[i]] = new_rk[sa[i - 1]] + (now != prev);
18C       }
14E       rk = new_rk;
4BE     }
201   }
00C   void count_sort() {
9A8     int n = sa.size();
7C4     vector<int> new_sa(n), cnt(n, 0), pos(n, 0);
CBF     for (auto e : rk) cnt[e]++;
3D7     for (int i = 1; i < n; i++) pos[i] = pos[i - 1] + cnt[i - 1];
1EC     for (auto e : sa) new_sa[pos[rk[e]]++] = e;
253     sa = new_sa;
A28   }
47E   void build_lcp() {
9A8     int n = sa.size();
75C     lcp.assign(n, 0);
617     for (int i = 0, h = 0; i < n; i++) {
F00       if (rk[i] == 0) {
33D         h = 0;

```

```

5E2     continue;
595     }
0A2     int j = sa[rk[i] - 1];
28E     while (s[i + h] == s[j + h]) h++;
CA3     lcp[rk[i]] = h;
254     if (h) h--;
7DF     }
2BE     }

// Intervalo [lo, hi] no SA onde p ocorre.
B04 pair<int, int> search(const string &p) const {
B63     int lo, hi, n = sa.size();
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
382             if (s.compare(sa[m], p.size(), p) < 0) l = m + 1;
EF3             else r = m;
103         }
0A8         lo = l;
CBD         }
F95         {
993             int l = 0, r = n;
40C             while (l < r) {
EE4                 int m = (l + r) / 2;
E02                 if (s.compare(sa[m], p.size(), p) <= 0) l = m + 1;
EF3                 else r = m;
D29             }
C1B             hi = l;
277         }
AAF         return {lo, hi};
EEF     }

// LCS; sep deve diferir de '$' e nao aparecer em a ou b.
4E7 static string lcs(const string &a, const string &b,
CCC     char sep = 1) {
B29     SuffixArray suf(a + sep + b);
37E     int m = a.size(), best = 0, pos = 0;
564     for (int i = 1; i < (int)suf.sa.size(); i++) {
935         bool in_a = suf.sa[i] < m;
A7D         bool prev_in_a = suf.sa[i - 1] < m;
B09         if (in_a != prev_in_a && suf.lcp[i] > best) {
F6B             best = suf.lcp[i];
2B8             pos = suf.sa[i];
92E         }
409     }
D5D     return suf.s.substr(pos, best);
514 }
019 };

```

Suffix Array (Linear)

```

/* Suffix Array O(N) – SA-IS (Induced Sorting)
 * Build O(N). Use quando N muito grande justificar o SA-IS.
 *
 * Campos publicos apos construcao:
 * sa[i] = indice do i-esimo sufixo em ordem lexicografica
 * rk[i] = rank do sufixo s[i..] (inverso de sa)
 * lcp[i] = tamanho do LCP entre sa[i-1] e sa[i]; lcp[0] = 0
 *
 * Metodos:
 * search(p) -> [lo, hi] no SA; hi-lo e o nº de ocorrencias
 */

```

```

FC8 struct SuffixArrayLinear {
AC0     string s;
58B     vector<int> sa, rk, lcp;
CAE     SuffixArrayLinear() {}

B7A     SuffixArrayLinear(const string &s) : s(s + '$') {
C75         vector<int> t(this->s.begin(), this->s.end());
C9A         sa = sa_is(t, 256);
9A8         int n = sa.size();
9BE         rk.resize(n);
665         for (int i = 0; i < n; i++) rk[sa[i]] = i;
E8F         build_lcp();

```

```

B40     }

B04 pair<int, int> search(const string &p) const {
B63     int lo, hi, n = sa.size();
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
382             if (s.compare(sa[m], p.size(), p) < 0) l = m + 1;
EF3             else r = m;
103         }
0A8         lo = l;
CBD         }
F95         {
993             int l = 0, r = n;
40C             while (l < r) {
EE4                 int m = (l + r) / 2;
E02                 if (s.compare(sa[m], p.size(), p) <= 0) l = m + 1;
EF3                 else r = m;
D29             }
C1B             hi = l;
277         }
AAF         return {lo, hi};
EEF     }

47E void build_lcp() {
9A8     int n = sa.size();
75C     lcp.assign(n, 0);
E74     for (int i = 0, h = 0; i < n - 1; i++) {
0A2         int j = sa[rk[i] - 1];
28E         while (s[i + h] == s[j + h]) h++;
CA3         lcp[rk[i]] = h;
254         if (h) h--;
09C     }
C83 }

67A template <typename T>
EE7 static vector<int> sa_is(const T &s, int sigma) {
163     int n = s.size();
979     if (n == 1) return {0};
4E1     if (n == 2)
32E         return s[0] < s[1] ? vector<int>{0, 1} : vector<int>{1, 0};

81A     vector<bool> t(n);
3A1     t[n - 1] = true;
9C8     for (int i = n - 2; i >= 0; i--)
537         t[i] = s[i] < s[i + 1] || (s[i] == s[i + 1] && t[i + 1]);

A15     auto is_lms = [&](int i) {
BCB         return i > 0 && t[i] && !t[i - 1];
0C4     };

FC6     vector<int> bkt(sigma + 1, 0);
7E5     for (int c : s) bkt[c + 1]++;
CA7     for (int i = 1; i <= sigma; i++) bkt[i] += bkt[i - 1];

CB5     auto induced_sort = [&](const vector<int> &lms) {
386         vector<int> sa(n, -1);
CDA         vector<int> b = bkt;
5EB         for (int i = (int)lms.size() - 1; i >= 0; i--)
F9D             sa[--b[s[lms[i]] + 1]] = lms[i];
332         b = bkt;
830         for (int i = 0; i < n; i++)
DC3             if (sa[i] > 0 && !t[sa[i] - 1])
420                 sa[b[s[sa[i] - 1]]++] = sa[i] - 1;
332         b = bkt;
056         for (int i = n - 1; i >= 0; i--)
C0D             if (sa[i] > 0 && t[sa[i] - 1])
3ED                 sa[--b[s[sa[i] - 1] + 1]] = sa[i] - 1;
DB7         return sa;
66E     };

021     vector<int> lms;
830     for (int i = 0; i < n; i++)
B39         if (is_lms(i)) lms.push_back(i);
EF0     vector<int> sa = induced_sort(lms);

835     vector<int> rk(n, -1);
8F1     int cls = 0;

```

```

975     for (int i = 0, prev = -1; i < n; i++) {
FAD         if (!is_lms(sa[i])) continue;
837         if (prev != -1) {
047             bool diff = false;
5B8             for (int d = 0;; d++) {
12C                 if (s[prev + d] != s[sa[i] + d] ||
A89                     t[prev + d] != t[sa[i] + d]) {
570                     diff = true;
C2B                     break;
36D                 }
199                 if (d > 0 && (is_lms(prev + d) || is_lms(sa[i] + d)))
C2B                     break;
0A4             }
265             if (diff) cls++;
142         }
7F0         rk[sa[i]] = cls;
C6D         prev = sa[i];
9FA     }

CDB     vector<int> lms2, rank2;
830     for (int i = 0; i < n; i++)
6A2         if (rk[i] != -1) {
D24             lms2.push_back(i);
213             rank2.push_back(rk[i]);
642         }

// Ha cls+1 classes distintas, numeradas de 0 a cls.
8F6     vector<int> sa2 = sa_is(rank2, cls + 1);
D14     vector<int> sorted_lms(lms2.size());
E14     for (int i = 0; i < (int)sa2.size(); i++)
42D         sorted_lms[i] = lms2[sa2[i]];

E0A     return induced_sort(sorted_lms);
50A }
160 };

```

Geometry

Point

```

107 const ld eps = 1e-9;
177 bool eq(ll a, ll b) {
605     return a == b;
6F1 }

D97 bool eq(ld a, ld b) {
BA0     return abs(a - b) <= eps;
BFC }

67A template <typename T>
F26 struct Point {
645     T x, y;
0FA     Point(T x = 0, T y = 0) : x(x), y(y) {}

C93     bool operator<(Point o) const {
AE1         return tie(x, y) < tie(o.x, o.y);
8D2     }
CE2     bool operator==(Point o) const {
651         return eq(x, o.x) && eq(y, o.y);
511     }
9E5     bool operator!=(Point o) const { return !(this == o); }

480     Point operator+(Point o) const { return {x + o.x, y + o.y}; }
229     Point operator-(Point o) const { return {x - o.x, y - o.y}; }
AE8     Point operator*(T k) const { return {x * k, y * k}; }
7C2     Point operator/(T k) const { return {x / k, y / k}; }

61A     T dot(Point o) const { return x * o.x + y * o.y; }
CB8     T cross(Point o) const { return x * o.y - y * o.x; }
// area com sinal do triangulo (this, a, b)
DBA     T cross(Point a, Point b) const {
491         return (a - *this).cross(b - *this);
61E     }
F68     T dist2() const { return x * x + y * y; }

AD2     ld len() const { return hypot((ld)x, (ld)y); }
001     Point<ld> norm() const {
DE8         ld l = len();
D97         return {x / l, y / l};

```

```

978 }
178 Point perp() const { return {-y, x}; } // rotacao 90° CCW
8CF Point<ld> rotate(ld a) const { // a em radianos
97E     return {x * cos(a) - y * sin(a), x * sin(a) + y * cos(a)};
CB4 }

539 friend ostream &operator<<(ostream &os, Point p) {
9B9     return os << p.x << " " << p.y;
DB4 }

2F4 };

29C using Pt = Point<ll>;
BA9 using Ptd = Point<ld>;

// angulo entre os vetores p e q
67A template <typename T>
AE1 ld angle(Point<T> p, Point<T> q) {
9AF     return atan2((ld)p.cross(q), (ld)p.dot(q));
447 }

// area com sinal do triangulo (p, q, r) - positivo = CCW
67A template <typename T>
D06 T sarea(Point<T> p, Point<T> q, Point<T> r) {
0CC     return p.cross(q, r);
B80 }

// se p, q, r sao colineares
67A template <typename T>
EB4 bool is_collinear(Point<T> p, Point<T> q, Point<T> r) {
B84     return eq(sarea(p, q, r), (T)0);
0D1 }

Line

C97 #include "point.hpp"

72C struct Line {
2DF     Ptd p, q;
2BD     Line() {}
299     Line(Ptd p, Ptd q) : p(p), q(q) {}
F5B };

// se p pertence ao segmento r
7E1 bool isinseg(Ptd p, Line r) {
AAC     Ptd a = r.p - p, b = r.q - p;
B7E     return eq(a.cross(b), 0) && a.dot(b) < eps;
C3B }

// projecao do ponto p na reta r
15B Ptd proj(Ptd p, Line r) {
BEA     if (r.p == r.q) return r.p;
2FF     Ptd v = r.q - r.p, u = p - r.p;
5C6     return r.p + v * (u.dot(v) / v.dot(v));
2E8 }

// Intersecao das retas; retorna {INF, INF} se paralelas.
24B Ptd inter(Line r, Line s) {
CAD     Ptd u = r.q - r.p, v = s.q - s.p, w = s.p - r.p;
018     ld d = u.cross(v);
F2F     if (eq(d, 0)) return {1e18, 1e18};
2C9     return r.p + u * (w.cross(v) / d);
15F }

// se o segmento r intersecta o segmento s
090 bool interseg(Line r, Line s) {
469     if (isinseg(r.p, s) || isinseg(r.q, s) || isinseg(s.p, r) ||
C1D         isinseg(s.q, r))
8A6         return true;
D02     auto ccw = [] (Ptd a, Ptd b, Ptd c) {
9B9         return (b - a).cross(c - a) > eps;
3ED     };
13C     return ccw(r.p, r.q, s.p) != ccw(r.p, r.q, s.q) &&
413         ccw(s.p, s.q, r.p) != ccw(s.p, s.q, r.q);
7D4 }

// distancia do ponto p a reta r
E49 ld disttoline(Ptd p, Line r) {
FF0     return abs((r.q - r.p).cross(p - r.p)) / (r.q - r.p).len();
2F5 }

// distancia do ponto p ao segmento r

```

```

A20 ld disttoseg(Ptd p, Line r) {
5A8     if ((r.q - r.p).dot(p - r.p) < 0) return (p - r.p).len();
A6E     if ((r.p - r.q).dot(p - r.q) < 0) return (p - r.q).len();
A19     return disttoline(p, r);
9C0 }

// distancia entre dois segmentos
68D ld distseg(Line a, Line b) {
4DF     if (interseg(a, b)) return 0;
3C3     return min({disttoseg(a.p, b), disttoseg(a.q, b),
458         disttoseg(b.p, a), disttoseg(b.q, a)});
A44 }

```

Circle

```

5FC #include "line.hpp"

// centro da circunferencia que passa por a, b, c
880 Ptd getcenter(Ptd a, Ptd b, Ptd c) {
3FB     Ptd m1 = (a + b) / 2, m2 = (a + c) / 2;
1E3     return inter(Line(m1, m1 + (b - a).perp()),
9A7         Line(m2, m2 + (c - a).perp()));
E56 }

// intersecao da circunferencia (c, r) com a reta ab
E6C vector<Ptd> circ_line_inter(Ptd a, Ptd b, Ptd c, ld r) {
5C2     vector<Ptd> points;
FAD     Ptd u = b - a, v = a - c;
F37     ld A = u.dot(u), B = v.dot(u), C = v.dot(v) - r * r;
1FA     ld D = B * B - A * C;
4AA     if (D < -eps) return points;
B40     points.push_back(c + v + u * (-B + sqrt(D + eps)) / A);
AFA     if (D > eps) points.push_back(c + v + u * (-B - sqrt(D)) / A);
97E     return points;
F33 }

// intersecao das circunferencias (a, r) e (b, R)
24A vector<Ptd> circ_inter(Ptd a, Ptd b, ld r, ld R) {
5C2     vector<Ptd> points;
8C7     ld d = (b - a).len();
701     if (d > r + R + eps || d + min(r, R) < max(r, R) - eps)
97E         return points;
398     ld x = (d * d - R * R + r * r) / (2 * d);
A5C     ld y = sqrt(max(0.0, r * r - x * x));
223     Ptd v = (b - a) / d;
EED     points.push_back(a + v * x + v.perp() * y);
526     if (y > eps) points.push_back(a + v * x - v.perp() * y);
97E     return points;
36B }

```

Polygon

```

5FC #include "line.hpp"

// 2x a area com sinal (>0 = CCW)
67A template <typename T>
CB6 T area2(const vector<Point<T>> &poly) {
D0C     T a = 0;
BC6     int n = poly.size();
830     for (int i = 0; i < n; i++)
F7A         a += poly[i].cross(poly[(i + 1) % n]);
3F5     return a;
1B9 }

// area do poligono
67A template <typename T>
402 ld polarea(const vector<Point<T>> &poly) {
323     return abs((ld)area2(poly)) / 2.0;
3C3 }

// 0 = fora, 1 = dentro, 2 = na borda
// 0(n) para qualquer poligono simples, convexo ou concavo.
67A template <typename T>
EA9 int winding(const vector<Point<T>> &poly, Point<T> p) {
C24     int n = poly.size(), w = 0;
603     for (int i = 0; i < n; i++) {
D79         Point<T> a = poly[i] - p, b = poly[(i + 1) % n] - p;
972         if (a.y <= 0 && b.y > 0) {
E31             T c = a.cross(b);

```

```

346         if (c > 0) w++;
EAB         else if (c == 0) return 2;
305     } else if (b.y <= 0 && a.y > 0) {
E31         T c = a.cross(b);
6EA         if (c < 0) w--;
EAB         else if (c == 0) return 2;
992     } else if (b.y == 0 && a.y == 0) {
09D         if (min(a.x, b.x) <= 0 && 0 <= max(a.x, b.x)) return 2;
88E     }
C79 }
2EA     return w != 0 ? 1 : 0;
E31 }

// Ponto no casco convexo CCW sem colineares - O(log n).
67A template <typename T>
406 bool inside_hull(const vector<Point<T>> &hull, Point<T> p) {
ACE     int n = hull.size();
D3E     if (n == 1) return p == hull[0];
4E1     if (n == 2)
DA7         return hull[0].cross(hull[1], p) == 0 &&
540             (p - hull[0]).dot(hull[1] - hull[0]) >= 0 &&
F0B             (p - hull[1]).dot(hull[0] - hull[1]) >= 0;
D16     if (hull[0].cross(hull[1], p) <= 0 ||
095         hull[0].cross(hull[n - 1], p) >= 0)
D1F         return false;
B9E     int lo = 1, hi = n - 1;
3D1     while (lo + 1 < hi) {
C86         int mid = (lo + hi) / 2;
353         if (hull[0].cross(hull[mid], p) > 0) lo = mid;
8C0         else hi = mid;
575     }
3B9     return hull[lo].cross(hull[lo + 1], p) >= 0;
FF3 }

// Corta pela reta r; mantem p quando (r.p, r.q, p) e CCW.
010 vector<Ptd> cut_polygon(vector<Ptd> poly, Line r) {
2D3     vector<Ptd> res;
BC6     int n = poly.size();
E89     auto ccw = [&](Ptd a, Ptd b, Ptd c) {
9B9         return (b - a).cross(c - a) > eps;
2B6     };
603     for (int i = 0; i < n; i++) {
D30         if (ccw(r.p, r.q, poly[i])) res.push_back(poly[i]);
C6F         if (n == 1) continue;
F59         Line s(poly[i], poly[(i + 1) % n]);
DEA         Ptd p = inter(r, s);
D7C         if (isinseg(p, s)) res.push_back(p);
C91     }
E88     res.erase(unique(res.begin(), res.end()), res.end());
EC2     if (res.size() > 1 && res.back() == res[0]) res.pop_back();
B50     return res;
DAD }

// se dois poligonos se intersectam - O(n*m)
B8C bool interpol(const vector<Ptd> &v1, const vector<Ptd> &v2) {
7D1     int n = v1.size(), m = v2.size();
68F     for (auto &p : v1)
E34         if (winding(v2, p)) return true;
043     for (auto &p : v2)
240         if (winding(v1, p)) return true;
830     for (int i = 0; i < n; i++)
A75         for (int j = 0; j < m; j++)
2B6             if (interseg(Line(v1[i], v1[(i + 1) % n]),
DBE                 Line(v2[j], v2[(j + 1) % m])))
8A6                 return true;
D1F     return false;
8F8 }

// Pick: I = (area2 - boundary + 2) / 2, boundary =
// Σgcd(|dx|, |dy|) nas arestas
5CC ll lattice_points(ll area2, ll boundary) {
4EE     return (area2 - boundary + 2) / 2;
4FB }

// distancia entre dois poligonos
CCD ld distpol(const vector<Ptd> &v1, const vector<Ptd> &v2) {

```

```

F6B  if (interpol(v1, v2)) return 0;
7D1  int n = v1.size(), m = v2.size();
D63  ld ret = le18;
830  for (int i = 0; i < n; i++)
A75      for (int j = 0; j < m; j++)
7DB      ret = min(ret, distseg(Line(v1[i], v1[(i + 1) % n]),
D87          Line(v2[j], v2[(j + 1) % m])));
EDF  return ret;
B9D }

```

Convex Hull

```

046 #include "polygon.hpp"

// Casco convexo CCW, O(N log N); descarta colineares.
67A template <typename T>
580 vector<Point<T>> convex_hull(vector<Point<T>> pts) {
CA0  int n = pts.size();
3C9  if (n < 2) return pts;
CD7  sort(pts.begin(), pts.end());
9C8  vector<Point<T>> h;
721  h.reserve(2 * n);
107  for (int phase = 0; phase < 2; phase++) {
5C9      int start = h.size();
D9F      for (auto &p : pts) {
C31          while ((int)h.size() >= start + 2) {
531              auto a = h[h.size() - 2], b = h.back();
9F6              if ((b - a).cross(p - a) > 0) break;
BFB              h.pop_back();
0ED          }
A49          h.push_back(p);
37A      }
BFB      h.pop_back();
05C      reverse(pts.begin(), pts.end());
0D0  }
E30  if (h.size() == 2 && h[0] == h[1]) h.pop_back();
81C  return h;
5AD }

// Operacoes O(log n) em casco CCW sem colineares.
6E2 struct ConvexPol {
1C4  vector<Ptd> pol;

C45  ConvexPol() {}
D90  ConvexPol(vector<Ptd> v) : pol(convex_hull(v)) {}

// se o ponto esta dentro do casco - O(log n)
46F bool is_inside(Ptd p) {
B1C  int n = pol.size();
03D  if (n == 0) return false;
E08  return inside_hull(pol, p);
66C }

// cmp(p, q) indica que p e mais extremo que q.
8AA int extreme(const function<bool(Ptd, Ptd)> &cmp) {
B1C  int n = pol.size();
4A2  auto extr = [&](int i, bool &cur_dir) {
22A      cur_dir = cmp(pol[(i + 1) % n], pol[i]);
35F      return !cur_dir && !cmp(pol[(i + n - 1) % n], pol[i]);
DDB  };
63D  bool last_dir, cur_dir;
A0D  if (extr(0, last_dir)) return 0;
993  int l = 0, r = n;
EAD  while (l + 1 < r) {
EE4      int m = (l + r) / 2;
F29      if (extr(m, cur_dir)) return m;
44A      bool rel_dir = cmp(pol[m], pol[l]);
72A      if (!!last_dir && cur_dir) ||
D60          (last_dir == cur_dir && rel_dir == cur_dir)) {
8A6          l = m;
1F1          last_dir = cur_dir;
69C      } else r = m;
D0E  }
792  return l;
B61 }

// Índice do ponto com maior produto interno com v - O(log n).
82A int max_dot(Ptd v) {

```

```

52C  return extreme(
30C      [&](Ptd p, Ptd q) { return p.dot(v) > q.dot(v); });
C6D  }

// Tangentes por p: indices {esquerda, direita} - O(log n).
D16 pair<int, int> tangents(Ptd p) {
2B1  auto left = [&](Ptd q, Ptd r) {
540      return (r - p).cross(q - p) > eps;
3F4  };
183  auto right = [&](Ptd q, Ptd r) {
F28      return (q - p).cross(r - p) > eps;
0FA  };
8EB  return {extreme(left), extreme(right)};
1AB  }
A34 };

```

Closest Pair

```

C97 #include "point.hpp"

/* Closest Pair - Par de pontos mais proximos
* Complexidade: O(N log N)
*
* closest_pair(pts) -> indices {i, j} do par; i < j.
*
* Distancia ao quadrado: (pts[i] - pts[j]).dist2()
* Funciona com coordenadas inteiras, sem erro de precisao.
*/

67A template <typename T>
CCC pair<int, int> closest_pair(const vector<Point<T>> &pts) {
CA0  int n = pts.size();
135  vector<int> idx(n);
295  iota(idx.begin(), idx.end(), 0);
658  sort(idx.begin(), idx.end(),
866      [&](int a, int b) { return pts[a].x < pts[b].x; });

3C9  T best = (pts[idx[0]] - pts[idx[1]]).dist2();
D94  pair<int, int> ans = {idx[0], idx[1]};

// set ordenado por (y, indice)
3B1  set<pair<T, int>> active;
637  for (int l = 0, r = 0; r < n; r++) {
BD5      const Point<T> &pr = pts[idx[r]];
FAF      T delta = (T)ceil(sqrt((double)best)) + 1;
3D6      while (pr.x - pts[idx[l]].x > delta) {
9CB          active.erase({pts[idx[l]].y, idx[l]});
63B          l++;
34B          delta = (T)ceil(sqrt((double)best)) + 1;
B99      }
4A9      auto lo = active.lower_bound({pr.y - delta, INT_MIN});
A87      auto hi = active.upper_bound({pr.y + delta, INT_MAX});
DC0      for (auto it = lo; it != hi; it++) {
6BD          T d = (pr - pts[it->second]).dist2();
1D6          if (d < best) {
365              best = d;
5A5              ans = {idx[r], it->second};
C22          }
D09      }
D2B      active.insert({pr.y, idx[r]});
FB8  }
591  if (ans.first > ans.second) swap(ans.first, ans.second);
BA7  return ans;
A71 }

```

Others

Stress Test

```

P=code # codigo a testar
Q=brute # brute correto
C=      # checker; vazio usa cmp

```

```

g++ "$P.cpp" -o sol -O2 || exit 1
g++ "$Q.cpp" -o brt -O2 || exit 1
g++ gen.cpp -o gen -O2 || exit 1

```

```

if [ -n "$C" ]; then
    g++ "$C.cpp" -o chk -O2 || exit 1
fi

```

```

for ((i = 1; ; i++)); do
    echo "$i"
    ./gen "$i" > in
    ./sol < in > out
    ./brt < in > out2
    if [ -n "$C" ]; then
        ./chk in out out2
    else
        cmp -s out out2
    fi
    if [ $? -ne 0 ]; then
        echo "-> entrada:"
        cat in
        echo "-> saida code:"
        cat out
        echo "-> saida brute:"
        cat out2
        break
    fi
done

Gen

42A #define endl '\n'

E72 auto seed =
415 chrono::steady_clock::now().time_since_epoch().count();
F1A mt19937 rng(seed);
// int x = rng();

463 int uniform(int l, int r) {
A7F  uniform_int_distribution<int> uid(l, r);
F54  return uid(rng);
D9E }

E8D int main() {
886  ios::sync_with_stdio(false);
B95  cin.tie(0);

BB3  return 0;
268 }

```

Checker

```

F3B int main(int argc, char *argv[]) {
4DE  ifstream in(argv[1]), out_sol(argv[2]), out_brt(argv[3]);

BB3  return 0; // 0 = AC, 1 = WA
231 }

```

Hash Function

0 hash de cada subseção é calculado com MD5 (3 chars) sobre o código sem comentários e sem espaços em branco. Usado para verificar integridade durante contest.

0 hash de uma linha com } é o hash de todo o código desde a { que a abriu.

Para verificar: copie o código, rode o verificador e compare o prefixo.

Hash Verifier

```

95B string getHash(string code) {
049  ofstream("z.cpp") << code;
FC0  system("g++ -E -P -dD -fpreprocessed ./z.cpp"
041      " | tr -d '[:space:]' | md5sum > sh");
DB3  ifstream("sh") >> code;
78A  return code.substr(0, 3);
497 }

E8D int main() {
E82  string line, context;
0EE  stack<string> blocks({""});
266  while (getline(cin, line)) {

```

```
063     context = line;
808     for (char c : line)
66E         if (c == '{') blocks.push("");
CF0         else if (c == '}')
19B             context = blocks.top() + line, blocks.pop();
661     cout << getHash(context) + " " + line << endl;
4F5     blocks.top() += context;
697 }
56A }
```

Tabelas de Referência

Tipos, Constantes & Complexidade

type	bits	max	fmt	valor	obs
int	32	$\approx 2.1 \times 10^9$	%d	12! $\approx$ 479001600	int
uint	32	$\approx 4.3 \times 10^9$	%u	20! $\approx$ 2.4 $\times$ 10 ¹⁸	ll
ll	64	$\approx 9.2 \times 10^{18}$	%lld	$\left(\frac{33}{16}\right) \approx 1.2 \times 10^9$	int
ull	64	$\approx 1.8 \times 10^{19}$	%llu	$\left(\frac{66}{33}\right) \approx 7.2 \times 10^{18}$	ll
double	64	10 ³⁰⁸ , $\varepsilon \approx 10^{-15}$	%lf		
long dbl	80	10 ⁴⁹³² , $\varepsilon \approx 10^{-18}$	%Lf		

<i>n</i>	$\pi(n)$	<i>n</i>	<i>O</i> viavel
10 ⁴	1229	10 ⁸	<i>O</i> (<i>n</i>)
10 ⁵	9592	10 ⁷	<i>O</i> (<i>n</i> log <i>n</i>)
10 ⁶	78498	5 $\times$ 10 ³	<i>O</i> (<i>n</i> ²)
10 ⁷	664579	500	<i>O</i> (<i>n</i> ³)
10 ⁸	5761455	22	<i>O</i> (2 ^{<i>n</i>})
10 ⁹	59847534	11	<i>O</i> (<i>n</i> !)

Combinatória

Somatórios

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = n^2 \frac{(n+1)^2}{4}$$
$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k n^{m+1-k}$$
$$\sum_{i=0}^n c^i = \frac{c^{n+1}-1}{c-1}, \quad \sum_{i=0}^{\infty} c^i = \frac{1}{1-c}, \quad |c| < 1$$
$$\sum_{i=0}^n i c^i = \frac{n c^{n+2} - (n+1) c^{n+1} + c}{(c-1)^2} + c, \quad \sum_{i=0}^{\infty} i c^i = \frac{c}{(1-c)^2}, \quad |c| < 1$$
$$H_n = \sum_{i=1}^n \frac{1}{i} \approx \ln n + 0.5772, \quad \ln n < H_n < \ln n + 1$$
$$\sum_{i=1}^n H_i = (n+1)H_n - n, \quad \sum_{i=1}^n i H_i = \frac{n(n+1)}{2} H_n - \frac{n(n-1)}{4}$$

Identidades Binomiais

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k} = \frac{n}{k} \binom{n-1}{k-1} = \binom{n-1}{k} + \binom{n-1}{k-1}$$
$$\binom{n}{m} \binom{m}{k} = \binom{n}{k} \binom{n-k}{m-k}, \quad \binom{n}{k} = (-1)^k \binom{k-n-1}{k}$$
$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k, \quad x^n - y^n = (x-y) \sum_{k=0}^{n-1} x^{n-1-k} y^k$$
$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$
$$\sum_{k=0}^n k \binom{n}{k} = n 2^{n-1}, \quad \sum_{k=0}^n k^2 \binom{n}{k} = n(n+1) 2^{n-2}$$
$$\sum_{k=0}^n \binom{r+k}{k} = \binom{r+n+1}{n}, \quad \sum_{k=0}^n \binom{k}{m} = \binom{n+1}{m+1}$$
$$\sum_{k=0}^n \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}, \quad \sum_{k=0}^n \binom{k}{k}^2 = \binom{2n}{n}$$
$$\sum_{k=0}^r (-1)^k \binom{n}{k} = (-1)^r \binom{n-1}{r}$$
$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k, \quad (1+x)^{-n} = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k$$

Inversao binomial: $f(n) = \sum_k \binom{n}{k} g(k) \Leftrightarrow g(n) = \sum_k (-1)^{n-k} \binom{n}{k} f(k)$

Contagem

Permutacoes com repeticoes $n_1, ..., n_r$: $\frac{n!}{n_1! \cdots n_r!}$
Permutacoes circulares: $(n-1)!$
Stars & Bars: $x_1 + \cdots + x_k = n, \quad x_i \geq 0$: $\binom{n+k-1}{k-1}$
Inclusao-Exclusao: $|\bigcup_{i=1}^n A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \cdots \pm |A_1 \cap \cdots \cap A_n|$

Sequências Especiais

Catalan: $C_n = \frac{1}{n+1} \binom{2n}{n}$; 1, 1, 2, 5, 14, 42, 132, 429, ...
Conta: arvores binarias com *n* nos, triangulacoes de (*n*+2)-gono, *n* pares de parenteses validos.

$$\textbf{Fibonacci: } F_0 = F_1 = 1, \quad F_{-i} = (-1)^{i-1} F_i$$
$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}$$
$$F_{i+1} F_{i-1} - F_i^2 = (-1)^i \quad (\text{Cassini})$$
$$F_{n+k} = F_k F_{n+1} + F_{k-1} F_n, \quad F_{2n} = F_n F_{n+1} + F_{n-1} F_n$$
$$\gcd(F_m, F_n) = F_{\gcd(m,n)}, \quad \sum_{k=1}^n F_k = F_{n+2} - 1, \quad \sum_{k=1}^n F_k^2 = F_n F_{n+1}$$
$$\begin{pmatrix} F_n & F_{n-1} \\ F_{n-1} & F_{n-2} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n$$

Representacao de Fibonacci: todo *n* tem representacao unica $n = F_{k_1} + \cdots + F_{k_m}, \quad k_i \geq k_{i+1} + 2, \quad k_m \geq 2.$

Stirling 1* especie $\left[\begin{smallmatrix} n \\ 1 \end{smallmatrix} \right]$: permutacoes de *n* com *k* ciclos.

$$\left[\begin{smallmatrix} n \\ 1 \end{smallmatrix} \right] = (n-1)!, \quad \left[\begin{smallmatrix} n \\ 2 \end{smallmatrix} \right] = (n-1)! H_{n-1}, \quad \left[\begin{smallmatrix} n \\ n \end{smallmatrix} \right] = 1$$
$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = (n-1) \left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right] + \left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right], \quad \sum_k \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = n!$$
$$x^{\overline{n}} = \sum_{k=0}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k, \quad \left[\begin{smallmatrix} n \\ -1 \end{smallmatrix} \right] = \binom{n}{2}$$

Stirling 2* especie $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$: particoes de *n* em *k* subconjuntos nao vazios.

$$\left\{ \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ n \end{smallmatrix} \right\} = 1, \quad \left\{ \begin{smallmatrix} n \\ 2 \end{smallmatrix} \right\} = 2^{n-1} - 1, \quad \left\{ \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\} = \left[\begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right] = \binom{n}{2}$$
$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = k \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

Bell $B_n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$: # particoes de {1,...,*n*} em subconjuntos nao vazios.
 $B_0 = 1, B_1 = 1, B_2 = 2, B_3 = 5, B_4 = 15, B_5 = 52, B_6 = 203$

Numeros de Euler $\left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle$: permutacoes de *n* com *k* ascendentes.

$$\left\langle \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\rangle = \left\langle \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\rangle = 1, \quad \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle = \left\langle \begin{smallmatrix} n \\ n-1-k \end{smallmatrix} \right\rangle, \quad \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle = (k+1) \left\langle \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\rangle + (n-k) \left\langle \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\rangle$$
$$\left\langle \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\rangle = 2^n - n - 1, \quad \left\langle \begin{smallmatrix} n \\ m \end{smallmatrix} \right\rangle = \sum_{k=0}^m (-1)^k \binom{n+1}{k} (m+1-k)^n$$
$$x^n = \sum_{k=0}^n \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle \binom{x+k}{n}$$

Derangements: $D_n = (n-1)(D_{n-1} + D_{n-2})$; $D_n \approx \frac{n!}{e}$

Josephus: $f(1, k) = 0, \quad f(n, k) = (f(n-1, k) + k) \bmod n$

Funções Geradoras

Serie ordinaria: $A(x) = \sum_{i \geq 0} a_i x^i$; **exponencial:** $A(x) = \sum_{i \geq 0} a_i \frac{x^i}{i!}$

$$\frac{1}{1-x} = \sum x^i, \quad e^x = \sum \frac{x^i}{i!}, \quad \ln(1+x) = \sum (-1)^{i+1} \frac{x^i}{i}$$
$$(1+x)^n = \sum \binom{n}{i} x^i, \quad \frac{x}{1-x-x^2} = \sum F_i x^i$$
$$\frac{1}{2x} (1 - \sqrt{1-4x}) = \sum C_i x^i, \quad \frac{1}{\sqrt{1-4x}} = \sum \binom{2i}{i} x^i$$

Convolucao: $A(x)B(x) = \sum_i \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i.$

Soma acumulada: $B(x) = \frac{1}{1-x} A(x)$ se $b_i = \sum_{j \leq i} a_j.$

Potências Fatoriais & Bernoulli

$$x^{\underline{n}} = x(x-1) \cdots (x-n+1), \quad x^{\overline{n}} = x(x+1) \cdots (x+n-1)$$
$$x^{\underline{n}} = \sum_{k=1}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] (-1)^{n-k} x^k, \quad x^{\overline{n}} = \sum_{k=1}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k$$

$$x^n = \sum_{k=1}^n \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^{\overline{k}}$$

Numeros de Bernoulli ($B_i = 0$ para *i* impar > 1):
 $B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_4 = -\frac{1}{30}, \quad B_6 = \frac{1}{42}, \quad B_8 = -\frac{1}{30}$

$$\frac{x}{e^x - 1} = \sum_{i \geq 0} B_i \frac{x^i}{i!}, \quad n! = \sqrt{2\pi n} \left(\frac{n}{e} \right)^n \left(1 + \Theta\left(\frac{1}{n}\right) \right)$$

Teoria dos Números

Euler Phi & Möbius

$\varphi(n)$: quantidade de inteiros em [1,*n*] coprimos com *n*.

$$\varphi(n) = n \prod_p \left(1 - \frac{1}{p} \right) = \prod_{i=1}^n p_i^{e_i-1} (p_i - 1)$$

$$\varphi(mn) = \varphi(m)\varphi(n) \quad (\gcd = 1), \quad \sum_{d \mid n} \varphi(d) = n$$

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$
$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^r n = p_1 \cdots p_r, & \sum_{d \mid n} \mu(d) = [n = 1] \\ 0 & p^2 \mid n \end{cases}$$

Inversão de Möbius: $g(n) = \sum_{d \mid n} f(d) \Leftrightarrow f(n) = \sum_{d \mid n} \mu\left(\frac{n}{d}\right) g(d)$

$$\frac{1}{\zeta(x)} = \sum_{i=1}^{\infty} \frac{\mu(i)}{i^x}, \quad \frac{\zeta(x-1)}{\zeta(x)} = \sum_{i=1}^{\infty} \frac{\varphi(i)}{i^x}, \quad \zeta^2(x) = \sum_{i=1}^{\infty} \frac{d(i)}{i^x}$$

GCD, Diofantinas & TCR

$\gcd(a, b) = \gcd(b, a \bmod b)$; $\text{lcm}(a, b) = ab / \gcd(a, b)$
Bezout: $ax + by = \gcd(a, b)$ sempre tem solução inteira. Em geral, $ax + by = c$ tem solução $\Leftrightarrow \gcd(a, b) \mid c.$
TCR: Sistema $x \equiv a_i \pmod{m_i}$ com *m_i* coprimos dois a dois tem solucao unica mod *M* = $\prod m_i$:

$$x = \sum_i a_i M_i (M_i^{-1} \bmod m_i), \quad M_i = \frac{M}{m_i}$$

Aritmetica Modular

Inverso modular: $a^{-1} \bmod m$ via $a^{m-2} \bmod m$ (requer *m* primo). Para todos os inversos 1...*n* de uma vez:
 $\text{inv}[1] = 1, \quad \text{inv}[i] = -(m/i) \cdot \text{inv}[m \bmod i] \bmod m$

Lucas: Para *p* primo, $\binom{n}{k} \bmod p = \binom{n \bmod p}{k \bmod p} \cdot \left(\left\lfloor \frac{n}{p} \right\rfloor \right) \bmod p$

Recorrências & Séries

Teorema Mestre

$T(n) = aT(\frac{n}{b}) + f(n), \quad a \geq 1, \quad b > 1$:
 $f(n) = O(n^{\log_b a - \epsilon}) \Rightarrow T(n) = \Theta(n^{\log_b a})$
 $f(n) = \Theta(n^{\log_b a}) \Rightarrow T(n) = \Theta(n^{\log_b a} \log n)$
 $f(n) = \Omega(n^{\log_b a + \epsilon}) \text{ e } af(\frac{n}{b}) \leq cf(n) \Rightarrow T(n) = \Theta(f(n))$

Séries de Taylor

$$e^x = \sum \frac{x^i}{i!}, \quad \ln(1+x) = \sum (-1)^{i+1} \frac{x^i}{i}, \quad \ln \frac{1}{1-x} = \sum \frac{x^i}{i}$$
$$\sin x = \sum (-1)^i \frac{x^{2i+1}}{(2i+1)!}, \quad \cos x = \sum (-1)^i \frac{x^{2i}}{(2i)!}$$
$$\arctan x = \sum (-1)^i \frac{x^{2i+1}}{2i+1}, \quad \frac{1}{(1-x)^{n+1}} = \sum \binom{i+n}{i} x^i$$

Teoria dos Grafos

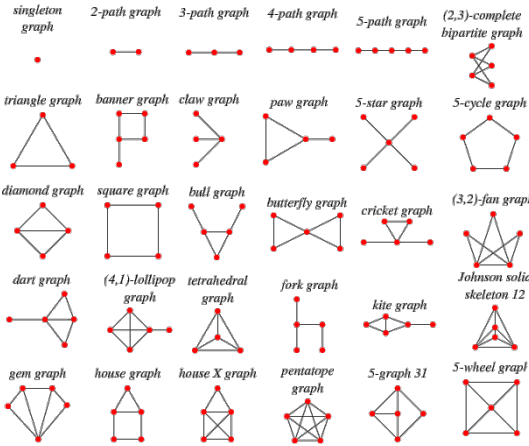
Teoremas

Euler: Circuito Euleriano $\Leftrightarrow$ todos os vertices tem grau par.
Caminho Euleriano $\Leftrightarrow$ exatamente 2 vertices de grau impar.
Fórmula de Euler (planar): $V - E + F = 2.$ Grafo planar simples: $E \leq 3V - 6$; planar bipartido: $E \leq 2V - 4.$ Todo planar tem algum vertice de grau $\leq 5.$
Cayley: Número de árvores geradoras rotuladas de *K_n* e *nⁿ⁻²*.
Kirchhoff: Número de árvores geradoras = qualquer cofator da matriz Laplaciana $L = D - A.$

Konig: Em grafo bipartido: cobertura mínima de vértices = emparelhamento máximo; conjunto independente máximo = V – emparelhamento máximo.
Hall: Existe emparelhamento perfeito de $U \Leftrightarrow \forall S \subseteq U: |N(S)| \geq |S|$.
Menger: Número máximo de caminhos aresta-disjuntos entre s e t = tamanho do corte mínimo s - t .
Dilworth: Em poset, partição mínima em cadeias = anti-cadeia máxima.
Mirsky: Em poset, partição mínima em anti-cadeias = comprimento da maior cadeia.
Prüfer: Bijecao entre árvores rotuladas de n vértices e sequências de comprimento $n - 2$ sobre $\{1, \dots, n\}$.
Erdos-Gallai: Sequência $d_1 \geq \dots \geq d_n$ é sequência de graus de algum grafo simples $\Leftrightarrow \sum d_i$ e par e para todo k :

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

Tipos de Grafo



Matemática

Probabilidade

Bayes: Atualiza a probabilidade de A dado que B ocorreu.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Quando $P(B)$ nao e conhecida diretamente (A_j sao hipoteses mutuamente exclusivas e exaustivas):

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

Esperanca: $E[X] = \sum xP(X = x)$

Linearidade: $E[aX + bY] = aE[X] + bE[Y]$. Se independentes: $E[XY] = E[X]E[Y]$

Variancia: $\text{Var}(X) = E[X^2] - E[X]^2$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Se independentes: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Markov: $P[X \geq \lambda E[X]] \leq \frac{1}{\lambda}$

Chebyshev: $P[|X - E[X]| \geq \lambda \sigma] \leq \frac{1}{\lambda^2}$

Binomial $B(n, p)$: n ensaios de Bernoulli independentes com prob p .

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad E[X] = np, \quad \text{Var} = np(1 - p)$$

Geometrica $\text{Geom}(p)$: numero de tentativas ate o primeiro sucesso ($q = 1 - p$).

$$P(X = k) = pq^{k-1}, \quad E[X] = \frac{1}{p}, \quad \text{Var} = \frac{q}{p^2}$$

Hipergeometrica $H(N, K, n)$: N itens, K marcados, n sorteados sem reposicao.

$$P(X = k) = \binom{K}{k} \frac{\binom{N-K}{n-k}}{\binom{N}{n}}, \quad E[X] = \frac{nK}{N}$$

Poisson $\text{Pois}(\lambda)$: aproximacao de $B(n, p)$ quando $n \rightarrow \infty, np = \lambda$.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad E[X] = \text{Var}(X) = \lambda$$

Normal $\mathcal{N}(\mu, \sigma^2)$: $p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$, $E[X] = \mu$, $\text{Var} = \sigma^2$.

Coupon collector: n tipos equiprovaveis; numero esperado de sorteios = nH_n .

Paradoxo do aniversario: $\approx 1.18\sqrt{M}$ amostras de $[M]$ para colisao com prob $\geq 50\%$.

Trigonometria

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

$$\sin 2x = 2 \sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$e^{ix} = \cos x + i \sin x, \quad e^{i\pi} = -1$$

Lei dos senos: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

Lei dos cossenos: $c^2 = a^2 + b^2 - 2ab \cos C$

Area do triangulo (lados a,b,c; angulos A,B,C):

$$\frac{1}{2} ab \sin C, \quad \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{a+b+c}{2}, \quad \frac{c^2 \sin A \sin B}{2 \sin C}$$

Circumraio: $R = ab \frac{c}{4A}$. **Inraio:** $r = \frac{A}{s}$.

Geometria

Shoelace: $A = \frac{1}{2} |\sum_i (x_i y_{i+1} - x_{i+1} y_i)|$

Pick: $A = I + \frac{B}{2} - 1$ (I = pontos interiores, B = pontos na borda)

Area por coordenadas: $\frac{1}{2} |\det \begin{pmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{pmatrix}|$

Esferas/circulos: $A = \pi r^2$, $V = \frac{4}{3} \pi r^3$.

Produto vetorial: $\vec{u} \times \vec{v} = (u_y v_z - u_z v_y, u_z v_x - u_x v_z, u_x v_y - u_y v_x)$

Em 2D: $u_x v_y - u_y v_x$ (> 0 anti-horario, $= 0$ colinear; modulo = area do paralelogramo)

Dist. ponto-ponto: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ (3D: inclui $(z_1 - z_2)^2$)

Dist. ponto-reta ($ax + by + c = 0$): $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

Rotacao 2D:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Rotacao 3D em torno de x, y, z por θ ($c = \cos \theta, s = \sin \theta$):

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{pmatrix}, \quad R_y = \begin{pmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{pmatrix}, \quad R_z = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Teoria dos Jogos

Sprague-Grundy: Todo jogo imparcial equivale a um Nim de um unico monte. O valor de Grundy (nimber) de uma posicao u e:

$$g(u) = \text{mex}\{g(v) : v \text{ alcançável de } u\}$$

onde $\text{mex}(S)$ e o menor inteiro nao negativo fora de S .

Posição perdedora $\Leftrightarrow g(u) = 0$. Em jogo composto de partes independentes:

$$g = g_1 \oplus g_2 \oplus \dots \oplus g_k$$

Nim: k montes de tamanhos $a_1, \dots, a_k$. Posição perdedora $\Leftrightarrow a_1 \oplus \dots \oplus a_k = 0$.

Nim misere (quem pegar o ultimo *perde*): subtraia 1 de cada pilha; se o XOR resultante $\neq 0$, a posicao original e ganhadora.

Wythoff: 2 montes (a, b) , $a \leq b$. Movimentos: remover qualquer qtd de um monte, ou a mesma qtd dos dois. Posição perdedora $\Leftrightarrow a = \lfloor \varphi k \rfloor$, $b = \lfloor \varphi^2 k \rfloor$ para algum $k \geq 0$, onde $\varphi = \frac{1+\sqrt{5}}{2}$.

Tabelas Numéricas

Números Primos

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73
79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181
191 193 197 199 211 223 227 229 233 239 241 251 257 263 269 271 277 281 283 293 307
311 313 317 331 337 347 349 353 359 367 373 379 383 389 397 401 409 419 421 431 433
439 443 449 457 461 463 467 479 487 491 499 503 509 521 523 541 547 557 563 569 571
577 587 593 599 601 607 613 617 619 631 641 643 647 653 659 661 673 677 683 691 701
709 719 727 733 739 743 751 757 761 769 773 787 797 809 811 821 823 827 829 839 853
857 859 863 877 881 883 887 907 911 919 929 937 941 947 953 967 971 977 983 991 997

Fibonacci

n	F_n	n	F_n	n	F_n	n	F_n	n	F_n	n	F_n	n	F_n
0	0	1	1	2	1	3	2	4	3	5	5	6	8
7	13	8	21	9	34	10	55	11	89	12	144	13	233
14	377	15	610	16	987	17	1597	18	2584	19	4181	20	6765

Triângulo de Pascal $\binom{n}{k}$

$n \setminus k$	0	1	2	3	4	5	6	7	8	9	10
0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

Debugging

```
-g++ -O2 -std=c++17 -Wall -Wextra  
-fsanitize=address,undefined  
-fno-sanitize-recover=all -g
```

Antes de submeter:

- Escreva casos simples se o sample não for suficiente.
- Limite de tempo apertado? Gere casos máximos.
- Memória está ok?
- Algo pode dar overflow?
- Está submetendo o arquivo certo?

Wrong Answer:

- Imprima sua solução e outputs de debug.
- Está limpando estruturas entre casos de teste?
- O algoritmo cobre todo o range de entrada?
- Releia o enunciado completo.
- Todos os casos de borda estão corretos?
- Entendeu o problema corretamente?
- Variáveis não inicializadas?
- Overflow?
- Confundindo N/M, i/j, etc.?
- Tem certeza que o algoritmo funciona?
- Que casos especiais você não considerou?
- As funções da STL funcionam como você acha?
- Adicione asserts e resubmeta.
- Rode em casos criados manualmente.
- Execute o algoritmo num caso simples à mão.
- Explique o algoritmo para um colega.
- Peça para o colega revisar seu código.
- Dê uma volta, vá ao banheiro.
- O formato de saída está correto? (espaços, newlines)
- Reescreva do zero ou peça ao colega.

Runtime Error:

- Testou todos os casos de borda localmente?
- Variáveis não inicializadas?
- Lendo/escrevendo fora do range de algum vetor?
- Algum assert que pode falhar?
- Divisão por zero? (mod 0?)
- Recursão infinita possível?
- Ponteiros ou iteradores invalidados?
- Usando memória demais?

Time Limit Exceeded:

- Tem loop infinito?
- Qual é a complexidade do algoritmo?
- Está copiando dados desnecessariamente? (use referências)
- Tamanho da entrada/saída? (considere scanf)
- Evite map/vector; use arrays/unordered_map.
- O que seus colegas acham do algoritmo?

Memory Limit Exceeded:

- Quanto de memória seu algoritmo realmente precisa?
- Está limpando estruturas entre casos de teste?